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INTERNATIONAL STANDARD

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Fibre optic communication subsystem test procedures – Part 4-4: Cable plants and links – Polarization mode dispersion measurement for installed links

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FIBRE OPTIC COMMUNICATION SUBSYSTEM TEST PROCEDURES –

Part 4-4: Cable plants and links – Polarization mode dispersion measurement for installed links

FOREWORD

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International Standard IEC 61280-4-4 has been prepared by subcommittee 86C: Fibre optic systems and active devices, of IEC technical committee 86: Fibre optics.

This second edition cancels and replaces the first edition published in 2006. This second edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

a) theory is removed and replaced with a reference to IEC TR 61282-9;
b) a new method, wavelength scanning OTDR and SOP analysis (WSOSA), is added as Annex G;
c) a brief description of each method is added to Clause 5;
d) Methods E and F are converted to informative Annexes E and F;

e) a new Clause (6) on measurement configurations is added;

f) a new Clause (7) on measurement considerations is added;

g) Clause 10 on procedure is expanded;

h) several of the apparatus diagrams are improved;

i) several clarifications about what is measured and what is calculated have been made in Annex H.

The text of this International Standard is based on the following documents:

CDV	Report on voting
86C/1378/CDV	86C/1419/RVC

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61280 series, published under the general title *Fibre optic communication subsystem test procedures*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

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INTRODUCTION

Polarization mode dispersion (PMD) is a statistical parameter. The reproducibility of measurements depends on the particular method, but is limited also by the PMD level of the link and the accessible wavelength range. Gisin [1][1] derived a theoretical limit to this reproducibility independent of the measurement method by assuming ideal measurement conditions.

Originally, the principles of IEC 61280-4-4:2006 were closely aligned with those of IEC 60793-1-48:2003 on optical fibre and optical fibre cable test method, which focuses on aspects related to the measurement of factory lengths. However, IEC 60793-1-48:2007 removed some of the test methods that are no longer of interest to fibre and cable manufacturers. These have been retained as informative Annexes D, E, and F in this document, and a new test method G has been added.

This document also updates test methods A, B and C and adds more information applicable to testing of installed cabling.

NOTE 1 Test methods for factory lengths of optical fibres and optical fibre cables are given in IEC 60793-1-48.

NOTE 2 Test methods for optical amplifiers (OAs) are given in IEC 61290-11-1 and IEC 61290-11-2.

NOTE 3 Test methods for passive optical components are given in IEC 61300-3-32.

NOTE 4 Guidelines for the calculation of PMD for links that include components such as dispersion compensators or optical amplifiers are given in IEC TR 61282-3.

NOTE 5 Further general guidance on PMD measurements and background theory is contained in IEC TR 61282-9.

[1] Figures in square brackets refer to the Bibliography.

FIBRE OPTIC COMMUNICATION SUBSYSTEM TEST PROCEDURES –

Part 4-4: Cable plants and links – Polarization mode dispersion measurement for installed links

1 Scope

This part of IEC 61280 provides uniform methods of measuring polarization mode dispersion (PMD) of single-mode installed links. An installed link is the optical path between transmitter and receiver, or a portion of that optical path. These measurements can be used to assess the suitability of a given link for high bit rate applications, or to provide insight on the relationships of various related transmission attributes. This document focuses on the measurement methods and requirements for measuring long lengths of installed cabling that can also include other optical elements, such as splices, connectors, amplifiers, chromatic dispersion compensating modules, dense wavelength division multiplexing or multiplexer (DWDM) components, multiplexers, wavelength selective switches, re-configurable optical add drop multiplexer (ROADMS).

This document focuses on the apparatus, procedures, and calculations needed to complete measurements. IEC TR 61282-9 explains the theory behind the test methods.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60793-1-44, *Optical fibres – Part 1-44: Measurement methods and test procedures – Cut-off wavelength*

IEC 61300-3-35, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 3-35: Examinations and measurements – Visual inspection of fibre optic connectors and fibre-stub transceivers*

IEC TR 61282-9, *Fibre optic communication system design guides – Part 9: Guidance on polarization mode dispersion measurements and theory*

IEC TR 62627-01, *Fibre optic interconnecting devices and passive components – Part 01: Fibre optic connector cleaning methods*

3 Terms, definitions, symbols and abbreviated terms

3.1 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/

- ISO Online browsing platform: available at http://www.iso.org/obp

3.2 Symbols and abbreviated terms

c	velocity of light in vacuum (299792458 m/s)
h	coupling length (also called h-parameter)
L	length of the link
t_c	optical source coherence time (Method C)
$\delta\lambda$	wavelength increment (interval, spacing or step size)
$\delta\nu$	optical frequency increment (interval, spacing or step size)
$\Delta\lambda$	optical source spectral width or linewidth (FWHM unless noted otherwise)
$\Delta\theta$	rotation angle on the Poincaré sphere
$\delta\tau$	differential arrival times of different polarization components
$\delta\tau_{\mathrm{max}}$	maximum measurable $\delta\tau$
$\delta\tau_{\mathrm{min}}$	minimum $\delta\tau$ value that can be measured
$\Delta\tau$	differential group delay value
$<\Delta\tau>$	average DGD over a wavelength range or PMD_{average} value
$<\Delta\tau^2>^{1/2}$	RMS DGD over a wavelength range or PMD_{RMS} value
$\Delta\tau_{\mathrm{max}}$	maximum $\Delta\tau$ value that can be measured
$\Delta\omega$	angular frequency variation in Method B
λ	test wavelength used to measure PMD
λ_0	central wavelength of the light source
ν	optical light frequency
σ_R	second moment of Fourier transform data
σ_0	RMS width of the squared autocorrelation envelope
σ_x	RMS width of the squared cross-correlation envelope
σ_ε	RMS width of interferogram
ω	angular optical frequency
ASE	amplified spontaneous emission
DCF	(chromatic) dispersion compensating fibre
DCM	(chromatic) dispersion compensating module
DGD	differential group delay
DOP	degree of polarization
DUT	device under test
DWDM	dense wavelength division multiplexing or multiplexer
FA	fixed analyzer
FA-FT	fixed analyzer-Fourier transform (PMD test method)
FBG	fibre Bragg grating
FET	field effect transistor
FWHM	full-width half-maximum
GINTY	general interferometric analysis (PMD test method)
INTY	interferometry (PMD test method)
I/O-SOP	input-output state of polarization
JME	Jones matrix eigenanalysis (PMD test method)

LED	light emitting diode
NLE	non-linear effect
OA	optical amplifier
OSA	optical spectrum analyzer
PDL	polarization dependent loss
PIN (diode)	positive intrinsic negative (diode layers)
PMD	polarization mode dispersion
PPS	polarization phase shift
PSA	Poincaré sphere analysis
PSP	principal SOP
RBW	resolution bandwidth
RMC	random mode coupling
RMS	root mean-square
ROADM	re-configurable optical add drop multiplexer
RTM	reference test method
SMF	single-mode fibre
SOP	state of polarization
SPE	Stokes parameter evaluation (PMD test method)
TINTY	traditional interferometric analysis (PMD test method)
WSOSA	wavelength scanning OTDR and SOP analysis (PMD test method)
WSS	wavelength selective switch

4 Background on PMD properties

PMD causes an optical pulse to spread in the time domain. This dispersion could impair the performance of a telecommunications system. The effect can be related to differential phase and group velocities and corresponding arrival times, $\delta\tau$, of different polarization components of the signal. For a sufficiently narrow band source, the effect can be related to a differential group delay (DGD), $\Delta\tau$, between pairs of orthogonally polarized principal states of polarization (PSP) at a given wavelength. For broadband transmission, the delays bifurcate and result in an output pulse that is spread out in the time domain. In this case, the spreading can be related to the root-mean square (RMS) of DGD values.

In long fibre spans, DGD varies randomly both in time and wavelength since it depends on the details of the birefringence along the entire fibre length. It is also sensitive to time-dependent temperature and mechanical perturbations on the fibre. For this reason, a useful way to characterize PMD in long fibres is in terms of an average DGD value over an appropriately large optical frequency range, either RMS $<\Delta\tau^2>^{1/2}$, the RMS DGD over this frequency range, or MEAN $<\Delta\tau>$, the (linear) mean of the DGD over this same frequency range. In principle, the average DGD value (RMS $<\Delta\tau^2>^{1/2}$ or MEAN $<\Delta\tau>$) does not undergo large changes for a given fibre from day to day or from source to source, unlike the parameters $\delta\tau$ or $\Delta\tau$. In addition, the average DGD value is a useful predictor of transmission performance.

The term PMD is used both in the general sense of two polarization modes having different group velocities (one having the fastest velocity and corresponding earliest arrival time and the other the slowest velocity and corresponding latest arrival time, the difference between the two arrival times being the DGD), and in the specific sense of the average DGD value (RMS $<\Delta\tau^2>^{1/2}$ or MEAN $<\Delta\tau>$). The latter gives us the strict definition of PMD for the purposes of this document. Although the DGD $\Delta\tau$ or pulse broadening $\Delta\delta$ is preferably averaged over frequency, for certain situations it can be averaged over time, or temperature. There are two metrics of averaged DGD, that is PMD:

$$PMD_{\text{MEAN}} = \langle \Delta\tau \rangle \tag{1}$$

$$PMD_{\text{RMS}} = \langle \Delta\tau^2 \rangle^{1/2} \tag{2}$$

The expression in Equation (1) is the PMD definition in term of the linear average of the DGD values. The expression in Equation (2) is the PMD definition in term of RMS average of the DGD values.

For many links, the DGD values are randomly distributed closely as a Maxwell distribution. Under the assumption of a perfect fit with a Maxwell distribution, the linkage between the two metrics, linear average DGD and RMS DGD is given by Equation (3).

$$< \Delta\tau > = \left(\frac{8}{3\pi}\right)^{1/2} < \Delta\tau^2 >^{1/2} \tag{3}$$

NOTE Equation (3) applies if the distribution of DGD values is Maxwellian. This assumption may not be valid if there are highly birefringent elements (relative to the rest of the link) in the optical path. A multiplier of 3 to 3,7 (see IEC TR 61282-3), depending on probability limits accepted by the link owner, is applied to the PMD_{AVG} value to determine the maximum DGD, which is specified for ITU-T compliant links. This multiplier is based on a Maxwell assumption and reflects a very long tail of that distribution. If the link includes a highly birefringent element, both the PMD_{AVG} and PMD_{RMS} metrics will increase relative to the actual tail of the DGD distribution (implying that a reduced multiplier could be used), but Equation (3) will not be maintained because the DGD distribution will begin to resemble one based on the square root of a non-central chi-square distribution with three degrees of freedom. In these cases, the PMD_{RMS} value will generally be larger relative to the PMD_{AVG} value indicated by Equation (3). This condition is indicated by “flat tops” on the fringe envelopes from the time domain measurement methods such as Method C and by bimodal DGD distributions from the frequency domain measurement methods such as Method B.

The expected value operator in the above equations refers to the long term expected value across all wavelengths. In practice, a finite wavelength range at a particular point of time and condition are sampled and some form of mean of the data is calculated. The expected value of these calculated means is equal to the long-term expected values, assuming ergodicity over time, wavelength, and condition. If this assumption is not valid, the result will vary depending on the particular wavelengths that are sampled. For ergodic conditions, the reproducibility of the measurement will vary with wavelength range and PMD level [1].

NOTE Ergodic: of or relating to a process in which every sequence or sizable sample is equally representative of the whole.

5 Measurement methods

5.1 Methods of measuring PMD

5.1.1 General

Seven basic methods of measuring PMD are given. Details specific to Methods A, B, C and G are given in normative annexes. Details specific to Methods D, E, and F are given in informative annexes. Methods A, B, C and G are in widespread commercial use and have been implemented in field test equipment. Methods A, B and C are also applicable to testing of fibres and cables in a factory environment as detailed in IEC 60793-1-48:2007. For some methods, multiple approaches of analyzing the measured results are also provided.

- Method A Fixed analyzer (FA)
  - Fourier transform (FT)
- Method B Stokes parameter evaluation (SPE)
  - Jones matrix eigenanalysis (JME)
  - Poincaré sphere analysis (PSA)

- Method C Interferometry (INTY)
  - Traditional analysis (TINTY)
  - General analysis (GINTY)
- Method D Stokes parameter evaluation using back-reflected light
  - Jones matrix eigenanalysis (JME)
  - Poincaré sphere analysis (PSA)
- Method E Modulated phase-shift
  - Full search
  - Mueller set analysis
- Method F Polarization phase shift (PPS)
- Method G Wavelength scanning OTDR and SOP analysis (WSOSA)
  - DGD analysis
  - PMD analysis

Each method has advantages in certain respects and disadvantages in others. These are discussed in more detail in Clause 7.

5.1.2 Method A: Fixed analyzer with Fourier transformation (FA-FT)

Method A requires only one input SOP from a broadband source or a tuneable laser source, depending upon the implementation. A fixed analyzer is used to track the relative change in power of one element of the Stokes vectors as a function of wavelength. When a Fourier transform is applied to this data, the result is a virtual half-interferogram, which is evaluated in a manner similar to Method C (TINTY).

5.1.3 Method B: Stokes parameters evaluation (SPE)

Method B measures $\Delta\tau(\omega)$ by measuring a polarimetric response to a change of narrowband light across a wavelength range. For each wavelength, three known and distinct SOPs such as for example linear SOPs at nominally 0°, 45° and 90° (orthogonal on the Poincaré sphere) shall be launched and the output Stokes vector that is transmitted through the link is measured.

The change of these Stokes vectors with angular optical frequency (wavelength), ω, and, with change in input SOP, yields the DGD as a function of wavelength through relationships that are based on the following definitions:

$$\frac{d\hat{s}(\omega)}{d\omega} = \hat{\Omega}(\omega) \times \hat{s}(\omega) \qquad (4)$$

$$\Delta\tau(\omega) = \left|\hat{\Omega}(\omega)\right| \qquad (5)$$

where

$\hat{s}$ is the output Stokes vector at angular optical frequency ω;

$\hat{\Omega}$ is the PSP vector, also called polarization dispersion vector (PDV) at angular optical frequency ω;

$\Delta\tau$ is the DGD at angular optical frequency ω.

The JME is completed by transforming the output Stokes vectors to Jones matrices [2], appropriate combination of the matrices at adjacent wavelengths, and calculation of the eigenvalues of the result followed by the application of an argument formula to obtain the DGD at the base frequency.

The PSA is completed by doing matrix algebra on the normalized output Stokes vectors to infer the output Stokes vector associated with circular birefringence at adjacent wavelengths, followed by the application of an arcsine formula to obtain the DGD at the base frequency.

For both JME and PSA, the average of the DGD values for a given wavelength range yields the PMD value for that range. The JME and PSA are mathematically equivalent. When using a narrowband source, the usual care should be taken to minimize optical reflections that could give rise to coherent interference effects.

In addition, in the case of a start-stop-measure system (by opposition to a swept system), the step size of the narrowband source shall be carefully selected in accordance with the expected DGD spectrum (DGD variation as a function of wavelength). More detail on step-size selection is given in Clause B.2. The linewidth of the narrowband source shall then be adjusted to the selected step size in accordance with the Nyquist criterion.

5.1.4 Method C: Interferometric

Method C is based on a broadband light source that is polarized at one or more known and distinct SOPs. The cross-correlation of the emerging electromagnetic field is determined by the interference pattern of the output light. The characterization of this pattern is done by computing the RMS width of the cross-correlation envelope of the interferogram for TINTY or the squared RMS widths of both the cross-correlation and autocorrelation envelopes of the interferogram for GINTY. These RMS widths yield the PMD value for the wavelength range associated with the source spectrum (the source bandwidth). Method C measures the PMD_{RMS} value only.

5.1.5 Method D: Stokes parameter evaluation using back-reflected light

Method D is in principle the same as Method B, but the output Stokes vectors that are reflected from the end of the link back to the source are measured to obtain DGD values. A modified formula that takes the reflection into account is used to calculate the PMD value for a given band. Since the reflection is required, the method is not suitable for links containing isolators (often found in optical amplifiers). Care should also be taken as in all other cases where a narrowband light source is used in order to avoid coherence interference effects and Fabry-Perot etalon effects that could show as false DGD. These effects produce delays that will be recorded in any case. Results reported from literature [12] have demonstrated that Method D is limited to distances of the order of 40 km, except if a reflector is used at the end of the plant to enhance the reflected light intensity.

5.1.6 Method E: Modulated phase-shift technique

Method E measures the phase difference between two or more input SOPs for a modulated narrowband light source. The full search technique requires launching an alternating pair of orthogonal input SOPs to obtain a given phase shift. The axis of this pair is varied across the entire Poincaré sphere to obtain the maximum phase shift. The maximum phase shift difference that is found is used to deduce the DGD at the centre wavelength of the source spectrum. The Mueller set technique [3] requires launching four SOPs per wavelength (source spectrum), representing a Mueller set. Analysis of the phase shifts associated with these SOPs yields a DGD value for the wavelength. When the measurement is conducted for a number of wavelengths in a wavelength range, the average of the DGD values yields the PMD value for that range.

5.1.7 Method F: Polarization phase shift (PPS)

Method F measures the phase differences between different known and distinct SOPs such as linear and orthogonal input SOPs – similar to Method E. It is different from Method E in that only two orthogonal input SOPs are required. However, because of this, results at two wavelengths are required to calculate one DGD value.

5.1.8 Method G: Wavelength scanning OTDR and SOP Analysis (WSOSA)

Method G uses an OTDR with a tuneable laser to launch two pulses of light at closely spaced wavelengths through a polarization scrambler into the link under test. The reflected pulses of light from the far end of the link under test pass through the polarization scrambler on their way back into the OTDR and a polarising beam splitter is used to direct the different polarization components to a pair of detectors. The difference between the signal levels at the two detectors is measured, and this difference is compared between the two wavelengths. This process can be repeated for many different input/output SOPs at each wavelength pair and at many different wavelength pairs to evaluate the PMD over a representative wavelength range.

5.2 Document structure

Information common to all seven methods is contained in Clauses 1 through 13, and requirements pertaining to each individual method appear in Annexes A, B, C, D, E, F and G, respectively.

5.3 Reference test method

There is no reference test method (RTM) because each method has its own particular optimal applications.

6 Measurement configurations

6.1 Passive cabling link

The link to be tested typically ranges from 10 km to 160 km, with most spans on a long haul network being typically no more than 80 km long, and with a loss less than 22 dB at 1 550 nm. These spans will normally be made up of many drums of cable, each 2 km to 10 km in length, spliced together end-to-end (see Figure 1).

Figure 1 – Typical passive cabling link configuration

This means that the link can be considered to have "random mode coupling" (RMC). It is also likely that fibres installed after the year 2000 (and many before that) will be "spun" to introduce additional mode coupling.

Normally, the fibres would be terminated at patch panels by splicing on pigtails to provide connector interfaces at optical distribution frames (ODFs) within buildings.

All test methods can be used successfully on this type of measurement configuration, provided that the implementation has sufficient dynamic range and that the PMD falls within the measurement range of the system. Typically, the total PMD of a link would lie between 0,1 ps (e.g., 10 km at 0,03 ps/√km) and 12 ps (e.g., 160 km at 1 ps/√km).

6.2 Link including amplifiers

A DWDM system can be operational over a link that includes amplifiers (see Figure 2). If the system is to be upgraded to operate at a higher data rate, for example, then the PMD of the entire link including the amplifiers might need to be tested.

Such a link will typically range between 100 km and 2 000 km for terrestrial systems and can extend up to 6 500 km for trans-Atlantic and 13 000 km for trans-Pacific undersea cables.

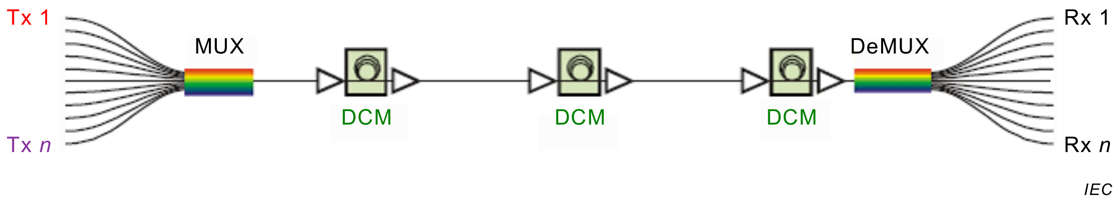


Figure 2 – Example configuration of link including amplifiers

With the exception of Methods D and G, all methods in this document can be used to measure the PMD in the gain band of links that include pumped optical amplifiers. Methods D and G rely on reflections coming back from the far end of the link under test. Amplifiers usually incorporate optical isolators that block any light travelling in the reverse direction.

6.3 Link including chromatic dispersion compensating modules

6.3.1 General

As shown in Figure 2, it is likely that a long haul link that contains amplifiers will also contain (chromatic) dispersion compensating modules.

A fibre based DCM will typically contain a long length (many kilometres) of dispersion compensating fibre (DCF) that has strongly negative chromatic dispersion to counteract the positive chromatic dispersion of a span of typical transmission fibre. This DCM can introduce significant amounts of PMD and therefore should be included in the measurement configuration.

Any of the test methods that can be used on amplified links can also be used on links that contain fibre based DCMs.

6.3.2 Grating based DCM

Increasingly, DCMs that use chirped fibre Bragg gratings are being used instead of fibre based DCMs. The applicability of the test methods to links that contain FBGs is for further study.

6.4 Link including ROADMs

6.4.1 General

Many networks are now built with re-configurable optical add drop multiplexers (ROADMs). These are network elements that allow the routing of different wavelengths in a DWDM system to different destinations whilst remaining in the optical domain. ROADMs are typically complex devices that include many functional elements, including amplifiers and wavelength selective switches (WSS), which result in tight optical filtering around the operational wavelengths. The presence of these elements can restrict the choice of test method. For example, the presence of amplifiers prevents the use of Methods D and G. The TINTY part of Method C requires a continuous spectrum and so cannot be used due to the filtering.

However, it may be possible to measure PMD and/or DGD in the following measurement configurations.

6.4.2 Multi-channel point to point configuration

If the ROADMs are set to carry multiple wavelengths between two locations, then Method A and the GINTY part of Method C can be used within the passbands of a set of wavelengths, and Method B can be used by setting a tuneable laser source to the DWDM wavelengths for which the ROADM is designed.

6.4.3 Single channel configuration

Depending upon the ROADM configuration, it may be possible to measure the DGD within the passband of a single channel, for example using either Method B or Method E.

7 Measurement considerations

7.1 General

When conducting PMD measurements on installed cabling, there are a number of considerations that can affect the choice of measurement technique and the accuracy of the results obtained. Clause 7 outlines some of these issues and their impact on the measurement. IEC TR 61282-9 shall be used to provide complete descriptions of the issues impacting the choice of measurement methods.

7.2 Wavelength range

In general, the broader the wavelength range of the measurement, the more accurate the result will be – known as the Gisin uncertainty [1].

Method E can yield DGD results for the narrowest occupied optical bandwidth of all the methods, but it requires the DGD of the link to be in excess of around 0,5 ps. The bandwidth requirement of Methods B, D and F is larger than that of Method E because numerical differentiation of the output data collected at two closely separated wavelengths is required, but these methods can measure DGD at the lowest of levels. Method A or Method C does not provide measurements of individual DGD values but does provide the RMS DGD (PMD_{RMS}) over the wavelengths of the source spectrum.

Method F is similar to Method E when Method E is implemented with the Mueller matrix approach. Method F, however, requires measurements at two adjacent wavelengths to obtain one DGD value, similar to Methods B and D.

7.3 PMD measurement range

Some of the test methods have limits on the upper or lower bound of the amount of PMD that they are capable of measuring, depending upon the particular implementation. For example, consider the following.

- The minimum PMD that can be measured using Method A is determined by the spectral range of the measurement to typically 0,1 ps, unless a DGD offset is used to remove this limitation.
- The maximum PMD that can be measured using Method A is determined by the spectral resolution of the detection system. If a high resolution spectrum analyzer is used, this limit can be 80 ps or more.

- The minimum PMD that be measured using Method C (TINTY) is determined by the spectral width of the source being used. The broader the source, the lower the PMD that can be measured. Typical limits are around 0,1 ps.
- The maximum PMD that can be measured using Method C is limited by the range of mirror movement in the interferometer. Typical limits are 100 s of ps.

7.4 Measurement dynamic range

It is necessary to ensure that the measurement system is capable of measuring over the expected loss of the link configuration. Single ended measurement techniques (D and G) that rely on a reflected signal from the far end of the link are particularly likely to suffer from a poor dynamic range. The performance of these systems can be improved by attaching a reflective termination to the far end of the link under test.

Where a broadband light source is used, it should have sufficient spectral power density for the measurement to be carried out.

Some measurement techniques (e.g., Method C) can have reduced dynamic range in the presence of larger amounts of PMD.

7.5 Fibre movement

Methods A, B, D, E, F and G may rely on measurements that can be disrupted if the link is vibrating, such as in the case of aerial cables, if the optical properties of the fibre change within the time used to measure the data for calculating individual DGD values. Fast measurement rates have been implemented in some commercial field test systems to reduce this effect. Methods A, B and G were demonstrated to give satisfactory performance in a field trial on a variety of installation environments [4], [5]. The data from this trial on the rate of change of the SOP showed that even underground cables could be subject to sudden transients in the SOP, and there were many environmental factors that could cause a change in the SOP, including vibration induced by the 50Hz of AC current on fibres attached to power line routes [6].

Method C can be used in the case of fibre movement, because it uses a broadband source and interferometry to deduce the PMD from the time-domain cross-correlation envelope of the measured interferogram.

7.6 Input and output SOP scrambling

7.6.1 General

The statistical basis of some of the PMD test methods (Methods A, C and G) can be improved by carrying out repeated measurements with a change in the input (and/or the output) state of polarization between measurements. This can help improve measurement certainty, particularly for low levels of PMD.

7.6.2 Polarizers/scramblers

Physically, SOP-scrambling consists of inserting controllable polarizers, one at the input and one at the output of the DUT, such that different input-SOPs and analyzer axes are set for a measurement scan. Multiple input/output SOPs can be selected to obtain a more complete set of measurements than would be possible for a single I/O-SOP setting. While a measurement obtained from a single setting is an option that would not require a scrambler, doing multiple settings will improve the precision of the result to a PMD value equal, for instance to the result from Method B. Multiple analyzer settings can also be used to obtain multiple settings (see C.2.1.3). Conceptually, the scrambling process can be viewed as follows:

- the input-polarizer followed by the scrambler acts as a single unit, an equivalent polarizer whose axis is set at any point on the Poincaré sphere in order to define the input-SOP;

- the scrambler followed by the analyzer acts as an equivalent analyzer whose axis is set at any point on the sphere in order to define the output-SOP.

One set of input-SOP/analyzer axis combinations will be labelled as one I/O-SOP. The aim is to get a set of measurements over uniformly distributed I/O-SOPs. In practice, there exists a number of possible ways to achieve this goal, as shown below.

7.6.3 The 9-states Mueller set

The sum of nine squared envelopes observed with nine specific I/O-SOPs is rigorously equal to the uniformly scrambled mean-squared envelope. These nine I/O-SOPs are three analyzer-axes forming a right-angled trihedron for each three input-SOPs also forming a right-angled trihedron.

7.6.4 Random scrambling

Random input/output SOP sampling has been implemented in various ways, including the following.

- Scan-to-scan scrambling: automatic/manual setting of the scramblers at each scan.
- Continuous scrambling: for some measurement methods including Method C, when squared envelopes are summed, scrambling can be performed while scanning. Automated scramblers are set to cover the sphere continuously as a function of time.
- Fast, single-scan scrambling: for some measurement methods including Method C, if scramblers are sufficiently fast, well-scrambled squared envelopes can be observed in a single-scan. However, this requires special provisions to avoid crosstalk between the AC part and the previously DC part of the interferogram.

7.7 Polarization dependent loss

None of the methods is suitable for measuring PMD of links with polarization dependent loss (PDL) in excess of 10 dB. Links with PDL values less than 1 dB can be measured with reasonable accuracy. Measurement accuracy can be compromised by the presence of PDL in excess of 1 dB.

7.8 Amplifier considerations

7.8.1 General

When the link to be tested includes optical amplifiers, there are several considerations; most of these relate to the presence of conventional Erbium-doped fibre amplifiers (EDFAs).

7.8.2 Optical isolators

Most amplifiers will incorporate optical isolators that block any light travelling in the reverse direction. This means that single ended test methods cannot be used.

7.8.3 Wavelength range

The amplifiers will only have a limited wavelength range of operation, commonly just the C-band (or sometimes the L-band). All measurements have to be made within this wavelength range. This can restrict some implementations of some of the test methods.

7.8.4 Power levels

It is important that the test signals are compatible with the intended operational power levels of the amplifiers in the link.

7.8.5 Amplified spontaneous emission (ASE) noise

When testing links including amplifiers, the amplified spontaneous emission (ASE) noise can generate depolarized spectral energy in the neighbourhood of the measurement wavelength. This will, in general, reduce the accuracy of the measurement. For Methods A, B, C, E and F, this effect can be moderated by implementing an optical or electrical filter at the receive end. However, optical filtering will not remove the ASE right under the signal spectrum. The accuracy will then be limited by a lower degree of polarization (DOP), if the spectral width of the filter cannot be sufficiently reduced as with a broadband source. Lower DOP can require the signal to be integrated longer to be meaningful, or the result will become too noisy, and interpretation will be erroneous.

7.9 Considerations on location of equipment

Using Method A and C, the source and receive equipment will typically be located at opposite ends of the link, but there is no need for active coordination of the source and receive equipment during the course of a measurement.

Methods B, E and F require a communications link and coordination between the source and receive equipment during a measurement, requiring an additional communication channel. The link itself can also be used for this channel.

Using Methods D and G, all measurement equipment is co-located at one end of the link under test, but special care is needed at the remote end to ensure that a suitable reflection can be obtained; this can require a reflective termination to be attached to the remote end.

8 Apparatus

8.1 General

The following apparatus is common to all measurement methods. Annexes A, B, C, D, E, F and G include layout drawings and other equipment requirements for each of the seven methods, respectively.

All methods require light sources that are controlled at one or more known and distinct states of polarization (SOP). All methods require injecting light across a spectral region to obtain a PMD value that is characteristic of that region (for example, C-band, C+L bands, S+C+L bands).

The methods differ in

- spectral characteristics of the source,
- bandwidth characteristics of the detection system,
- physical characteristics that are actually measured, and
- analysis methods.

All the transmission elements of the apparatus can contribute to the measured PMD. It is essential that these elements be selected for their lowest maximum DGD and PDL in the same spectral range of the PMD measurement. It may be necessary to assess the overall contribution of these elements by performing a measurement on a short length (2 m) of low PMD fibre. When the PMD contribution of these elements is significant, one option is to subtract, in quadrature, the measurement system PMD value from the measured link PMD value before reporting a final value.

8.2 Light source and polarizers

See Annexes A, B, C, D, E, F and G for detailed options of the spectral characteristics of the light source. The source shall produce sufficient radiation at the intended wavelength(s) and be stable in intensity over a time period sufficient to perform the measurement and should cover the required range of wavelengths for PMD determination for the band of interest.

For all methods, the light source is required to be polarized at one or more known and distinct SOPs before it is injected into the link under test. Polarizers, polarization adjusters using waveplates, liquid crystal retardation plates, loops of birefringent fibre that are mechanically moved, or electro-optic crystal devices can be used to set the source SOP at the input of the link under test as well as the analyzer polarization axis at the output of the link under test. The performance of the polarization adjuster setup can be verified by measuring the output power at three linear known and distinct SOPs. If the output powers are within 3 dB of one another, the adjuster setup is suitable.

For Methods A, B, D (when implemented using a narrowband tuneable light source – TLS), E, F and G, the effective spectral width (Gaussian spectrum, full-width half-maximum – FWHM) of the source, $\Delta\lambda$, shall be narrow enough so that light emerging from the link remains polarized under all conditions of measurement and for all DGD values being measured. A DOP of 90 % or greater is preferred, although measurements can be performed with values as low as 25 % with increased uncertainty. The relationship between DOP, the spectral width, $\Delta\lambda$, centre wavelength, λ_0, and DGD, $\Delta\tau$, is given by:

$$DOP(\%) = 100 \exp\left[-\frac{1}{4\ln(2)}\left(\frac{\pi c \Delta\tau \Delta\lambda}{\lambda_0^2}\right)^2\right] \tag{6}$$

8.3 Input optics

A connectorized test cord can be employed to launch the light into the link. The rest of the input optics precedes this test cord. The connection shall be stable over the course of the measurement. If the measurement technique can be affected by a change in the SOP during the measurement, then the test cord may need to be secured so as to prevent movement.

For Methods D and G, there are special requirements on interface of measurement system test cord and the link to control reflections (see Annexes D and G).

8.4 Cladding mode stripper

Use a device that extracts cladding modes from the link fibre. Under most circumstances, the fibre coating will perform this function.

8.5 High-order mode filter

Use a means to remove high-order propagating modes in the desired wavelength range that is greater than or equal to the cut-off wavelength of the link. IEC 60793-1-44 defines the cut-off wavelength measurement of single-mode fibre. For example, a one-turn bend of radius of 30 mm on the fibre is generally sufficient.

8.6 Output connection

A connectorized test cord can be employed to connect the link to the output optics. This connection shall be stable over the course of the measurement. If the measurement technique can be affected by a change in the SOP during the measurement, then the test cord may need to be secured so as to prevent movement.

For Methods D and G, the light reflected from the far end of the link (relative to the light source) may require a special end preparation to maximize the reflection. For this method, the output of the connection is the same as the input connection.

8.7 Output optics

See the appropriate annex (A, B, C, D, E, F or G).

Some link measurements can require the use of the combination of an optical amplifier and variable attenuator in the test set-up to maintain power at a level that is suitable for the analyzer detector. In this case, such power adjustment apparatus is generally placed immediately before the detector.

For Methods D and G, the output optics are co-located with the input optics.

8.8 Detector

For signal detection, an optical detector is used which is linear and stable over the range of intensities, wavelengths and measurement times that are encountered in performing the measurement. A typical system can include synchronous detection by a chopper/lock-in amplifier, optical power meter, optical spectrum analyzer, interferometer, or polarimeter. To use the entire spectral range of the source, the detection system shall have a wavelength range, which includes the wavelengths produced by the light source. See the appropriate annex (A, B, C, D, E, F or G) for additional details.

8.9 Computer or test platform

Use a computer or appropriate test platform to perform operations such as controlling the apparatus, taking intensity measurements, and processing the data to obtain the final results.

8.10 Means to reduce the effects of amplified spontaneous emission

ASE results in the depolarization of the received light. Depending on the implementation and method, the following alternatives can be used:

- modulation of the signal light in combination with an electrical filter within the detection electronics;
- a tuneable optical filter locked to the wavelength of the signal, placed at the output of the fibre link.

9 Sampling and specimens

A specimen is one of the link configurations listed in Clause 6. Except for Method C, the link should be measured at a time when vibrations or temperature variations are minimized over the time period of the measurement. It may be necessary for the extent of any change in the SOP to be monitored to verify that conditions are sufficiently stable for the test system.

If the link includes optical amplifiers, ROADMs, or other active devices, they should be turned on.

10 Procedure

a) Prepare the ends of the link by inspecting the connector end faces in accordance with IEC 61300-3-35 and by cleaning, if required, following the recommendation of IEC TR 62627-01. For Methods D and G, ensure that there is a strong enough reflection from the far end of the cabling under test and, if required, attach a reflective termination. For Method D, an angled fibre end is required at the input to minimize reflections from the input end.

b) Attach the ends to the input and output optics.

NOTE For Methods D and G, the input and output optics are the same; the end of the link to which they are attached will be called the input end.

c) Verify that there is enough signal strength present for the measurement.
d) Verify that the signal is sufficiently polarised.
e) Verify that the SOP is sufficiently stable for the measurement to be carried out.
f) Engage the computer or test platform to complete the scans and measurements found in Annexes A, B, C, D, E, F or G for the seven measurement methods.
g) Verify that the measurement scan is valid and exhibits the expected characteristics.
h) Store the measurement scan and complete the required documentation.
i) If required to improve the statistical basis of a measurement set, change the input and output state of polarization and repeat the measurement as many times as required to give the required measurement uncertainty.

11 Calculation or interpretation of results

Annexes A, B, C, D, E, F and G provide calculations to convert the measured data into DGD or PMD values.

IEC TR 61282-3 provides information on the calculation of the link PMD based on the statistics of optical fibre and component values.

12 Documentation

12.1 Information required for each measurement

– Link:
  - identification;
  - description, for example presence of optical amplifiers or other components;
  - length (km);
  - fibre and cable types.
– Testing date
– Test results:
  - PMD value as PMD_{AVG} or PMD_{RMS};
  - spectrum and linewidth of the test set-up light source;
  - DGD as a function of wavelengths or/and frequencies for Methods B, D, E, F;
  - interferogram for Method C;
  - intensity as a function of wavelengths or frequencies and Fourier transform for Method A;
  - for WSOSA – the detail of the wavelengths and wavelength increments used for the measurement.
– Wavelength or frequency range over which the measurement is performed.

12.2 Information to be available

The following information will normally be traceable by recording the make, model, serial number, and relevant module details of the equipment used.

– Test set-up:
  - measurement method;
  - calculation approach (analysis);
  - description of equipment:
    – source detailed characteristics;

  - analyzer detailed characteristics;
  - any other accessories if used.
- calibration:
  - date of latest calibration;
  - method of calibration;
  - uncertainty analysis.

– For Methods B, D, E and G, the number of wavelengths sampled.

– For Method C, the type of fringe detection technique.

13 Specification information

– Wavelength range.

– Any deviations from this procedure.

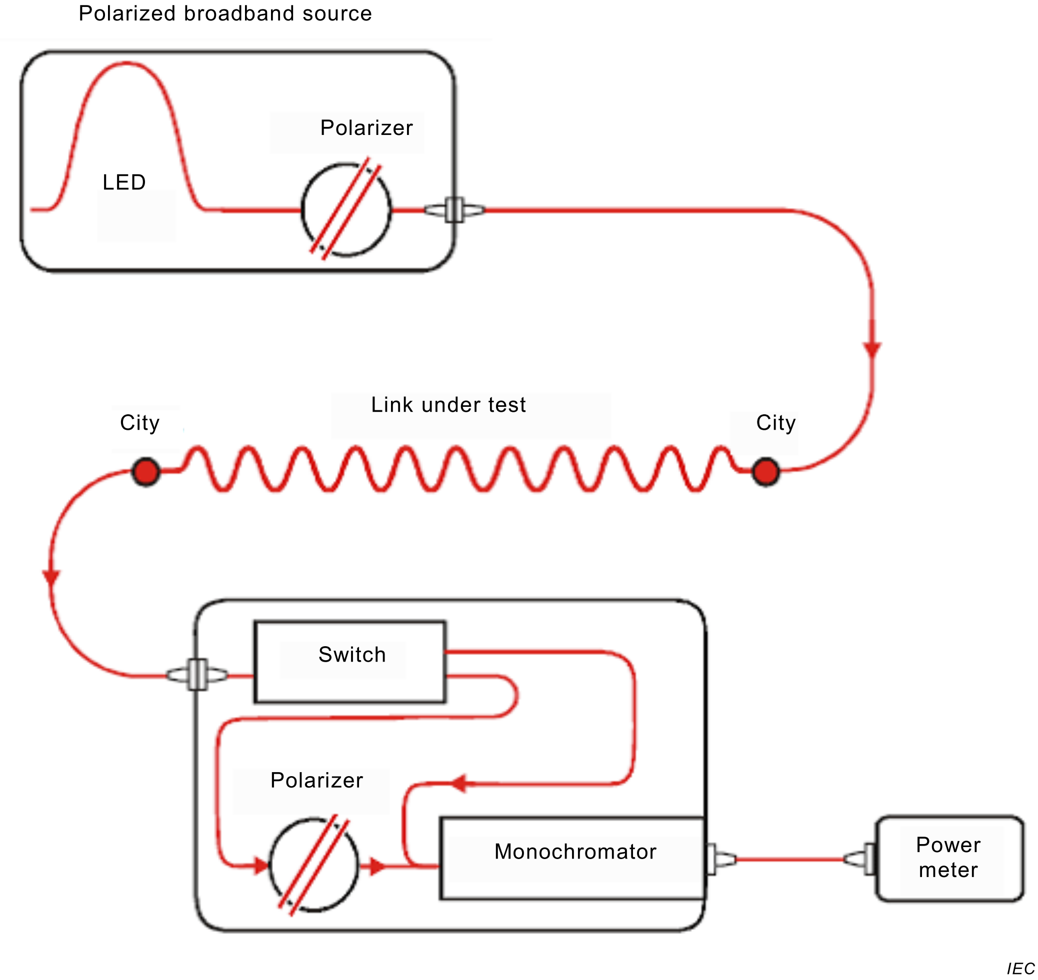
Annex A
(normative)

Fixed analyzer method

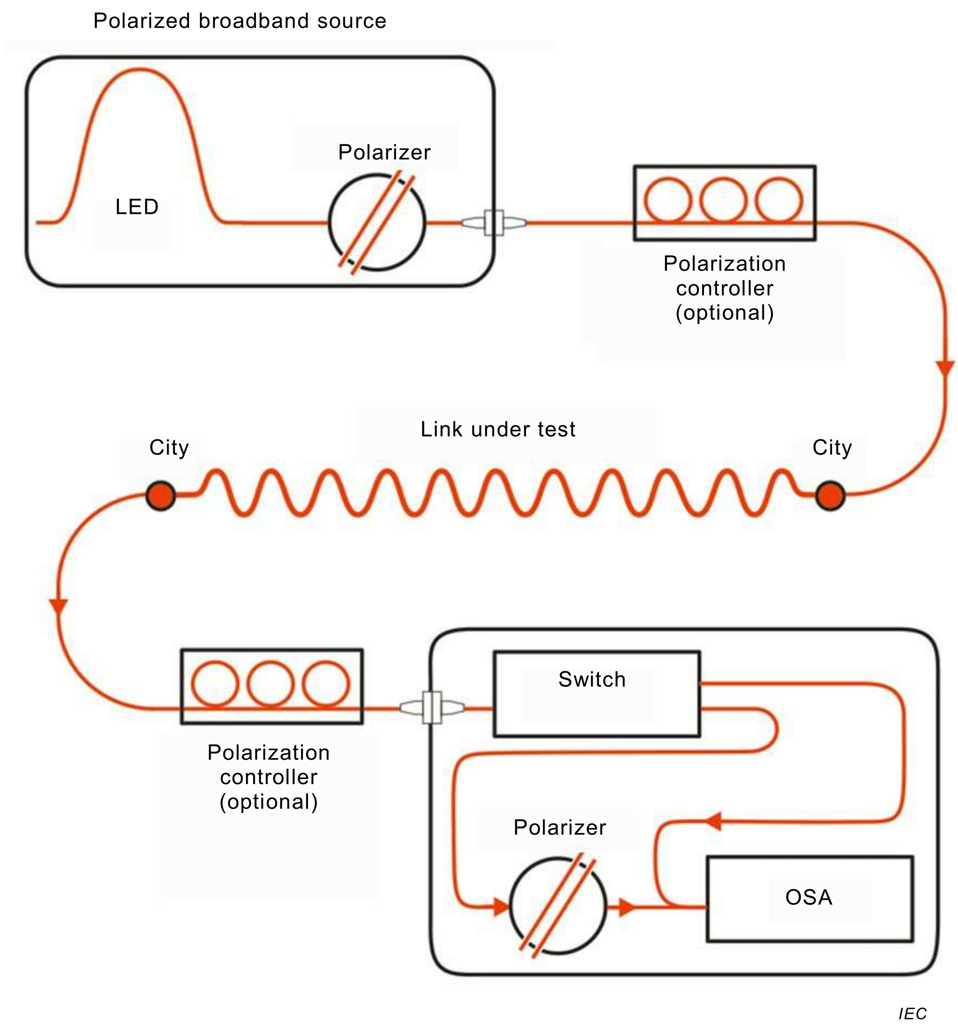
A.1 Apparatus

A.1.1 Block diagrams

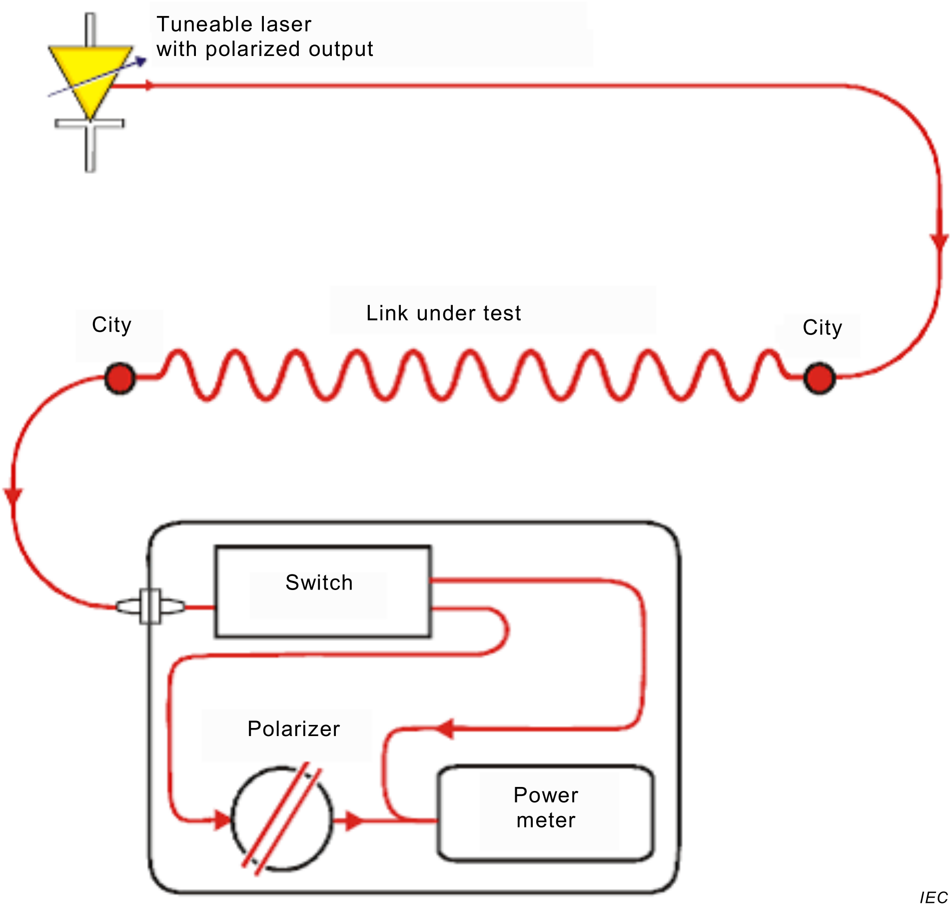
Figure A.1 shows possible block diagrams.



(a) Set-up using a broadband source and monochromator



(b) Set-up using a broadband source and optical spectrum analyzer (OSA)



(c) Set-up using a tuneable laser source

Figure A.1 – Block diagrams for fixed analyzer

A.1.2 Light source

In all cases, two kinds of light sources can be used, broadband or narrowband, depending on the type of analyzer. A broadband source such as the LED and polarizer combination shown in Figure A.1(a) and Figure A.1(b) can be used; an ASE source can be used to replace the LED. Alternatively, a narrow-band tuneable laser source can be used, as shown in Figure A.1(c).

Note that the measurement technique involves a reference scan of the power levels present as a function of wavelength, so the shape of the spectral content of the source has no effect on the measurement. There is no requirement for a continuous spectrum, so this technique can also be applied to systems containing optical filters (such as ROADMs).

When a broadband source is used, wavelength selection can be carried out using a narrow bandpass filtering analyzer such as an optical spectrum analyzer, a monochromator or an interferometer used as a Fourier transform spectrum analyzer placed before the analyzer. In the case of broadband source, the width of the filter is taken as the spectral width for the purpose of calculations.

In both cases, the spectral width shall be sufficiently small to maintain the desired DOP (see 8.2), but not too small compared to the selected step size to avoid unnecessary coherent interference effects and other spurious noises. In both cases, the range of wavelengths shall be sufficient to provide a PMD measurement of sufficient precision at the specified wavelength region (see A.3).

To insure that all features in the optical spectrum are adequately resolved, the spectral width should satisfy:

$$\Delta\lambda / \lambda_0 < \left(8\upsilon \Delta\tau_{max}\right)^{-1} \tag{A.1}$$

where

$\nu = c/\lambda$ is the optical frequency;

$\Delta\lambda$ is the spectral width;

$\Delta\tau_{max}$ is the maximum anticipated DGD.

For λ in the vicinity of 1 550 nm, Equation (A.1) reduces to the condition that $\Delta\lambda$ (nm) should be less than the reciprocal of $\Delta\tau$ (ps).

A.1.3 Analyzer

The angular orientation of the analyzer is not critical but should remain fixed throughout the measurement. With negligible mode coupling or low PMD values, some adjustment of the analyzer can be helpful in maximizing the amplitude of the oscillations in Figure A.2. This can also be achieved by manipulating the test cords at the input or output to adjust the state of polarization of the signal entering the receiver.

NOTE The analyzer can be replaced by a polarimeter.

A.1.4 Optional polarization control at the input and output of the link under test

It is an option to insert polarization controllers at the input and output of the link under test. The precision of the measurement will be improved when averaging multiple measurements performed for various polarization launch conditions at the input and output. The polarization changes can be either random or in a controlled sequence, the important factor being the achievement of a representative set of polarization states on the Poincaré sphere. The polarization states can be modified in between two spectral scans or be continuously scanned at a rate sufficiently slow compared to the spectral scan speed.

Note that it is also possible to generate a polarization rotation over a given wavelength span by using a Lyot type depolarizer at the input. The periodic polarization oscillations induced by the Lyot depolarizer can be demodulated in the time domain, which is straightforward with the Fourier transform of the optical spectrum or on an interferogram. Provided the polarization covering is sufficient over the wavelength span considered, the Lyot depolarizer efficiently replaces the input polarization controller and avoids multiple input states, therefore strongly reducing the acquisition time.

A.2 Procedure

A.2.1 Wavelength range and increment

The procedure requires measuring the power as a function of wavelengths (or optical frequencies) over a range at a defined wavelength increment, once with the analyzer in the optical path and once without. The wavelength range can influence the precision of the result (see Clause A.3). The wavelength increment should be selected to satisfy Equation (A.1), with the wavelength increment replacing $\Delta\lambda$.

If the Fourier transform method is used, the wavelength increment should ideally be uniform in optical frequency, and the number of steps should be a power of 2. The monochromator step-size, expressed in optical frequency, $\delta\nu$, shall be a factor of two smaller than the "oscillation frequency" corresponding to the maximum DGD measured. Because of the large amount of power outside the second moment for highly mode-coupled fibres, the Nyquist condition requires a frequency increment smaller than three times the frequency of the RMS width (second moment) associated with the maximum anticipated DGD. That is:

$$\frac{1}{6\delta\nu} > \Delta\tau_{\text{max}} \tag{A.2}$$

NOTE The source spectral width is generally equal to or less than the smallest wavelength increment. For example, for $\Delta\tau_{\text{max}}$ = 0,67 ps, a monochromator spectral width of 2 nm at 1 550 nm ($\delta\nu$ = 249 GHz) is typical.

If from the Fourier transform it is evident that there is significant energy near $\Delta\tau_{\text{max}}$, the measurement should be repeated with a reduced increment.

A.2.2 Complete the scans

Complete the scan with the analyzer in the light path. Record the received power as $P_{\text{A}}(\lambda)$.

Remove the analyzer from the light path and repeat the scan. Record the total received power as $P_{\text{Tot}}(\lambda)$.

NOTE 1 Alternatively, the measurement can be completed on a point by point basis by recording the power levels with and without the analyzer for each wavelength and then moving on to the next wavelength.

NOTE 2 Fast scanning systems such as an optical spectrum analyzer will reduce the impact of fibre movement, causing a change in the state of polarization on the measurement. Fast scanning systems can be used successfully on aerial cables or in other installed environments where fibre movement can be observed. [4], [5].

Calculate the power ratio, $R(\lambda)$ as follows:

$$R(\lambda) = \frac{P_{\text{A}}(\lambda)}{P_{\text{Tot}}(\lambda)} \tag{A.3}$$

Figure A.2 shows an example of a randomly mode-coupled result typical of a link application.

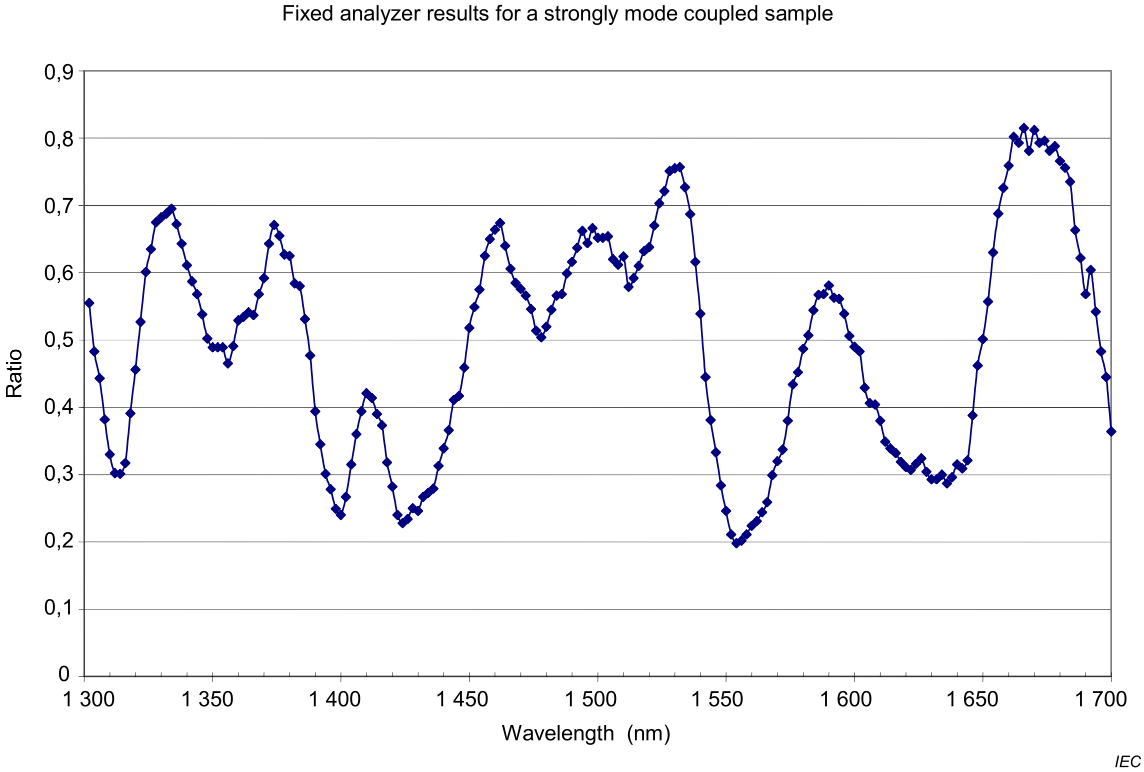


Figure A.2 – Example of the R-function for the fixed analyzer method

An alternative procedure is to leave the analyzer in place on the second scan, but rotate it 90°. Record the power as $P_B(\lambda)$. The formula for the power ratio is then:

$$R(\lambda) = \frac{P_A(\lambda)}{P_A(\lambda) + P_B(\lambda)} \quad \text{(A.4)}$$

NOTE 1 The cosine Fourier transform of the difference, $P_B - P_A$, divided by the sum is the cross-correlation function.

NOTE 2 If a polarimeter is used as the detection element, the normalized Stokes parameters are measured versus wavelength. The three spectral functions (one per vector element) are independent of received power and correspond to three independent difference functions between orthogonal analyzer states that can be analyzed in the same way.

A.3 Calculations – Fourier transform

A.3.1 General

A Fourier analysis of $R(\lambda)$, usually expressed in the domain of optical frequency ν, is used to derive PMD. The Fourier transform transforms this optical frequency domain data to the time domain. The Fourier transform yields direct information on the distribution of light arrival times $\delta\tau$. These data are post-processed as described below to derive the expected PMD, $<\Delta\tau>$, for the link under test. This method is applicable to links typically with random mode coupling. The method is, however, also applicable to negligible mode coupling.

Two additional analysis techniques are outlined in IEC TR 61282-9. These are cosine transform and spectral differentiation. These techniques would yield results that are mathematically equivalent to GINTY (Annex C). These techniques are not included in this document because there are no known commercial implementations.

A.3.2 Data pre-processing and Fourier transformation

To use this method, the Fourier transform normally requires equal intervals in optical frequency so that $R(\lambda)$ data are collected at λ values such that they form equal intervals in the optical frequency domain. Alternatively, data taken at equal λ intervals can be fitted (for example, by using a cubic spline fit) and interpolation used to generate these points, or more advanced spectral estimation techniques can be used. In each instance, the ratio $R(\lambda)$ at each λ value used is calculated using Equation (A.3) or Equation (A.4), as appropriate.

Zero-padding or data interpolation and DC level removal can be performed on the ratio data, $R(\lambda)$. Windowing the data may also be used as a pre-conditioning step before the Fourier transform. The Fourier transformation is now carried out to yield the amplitude data distribution $P(\delta\tau)$ for each value of $\delta\tau$.

A.3.3 Transform data fitting

A.3.3.1 General

Fourier transform data at zero $\delta\tau$ is due to the data having a non-zero average, which is sometimes referred to as direct current, or DC. Unless carefully removed, "DC" components in $R(\lambda)$ may be partially due to insertion loss of the analyzer, for example. When the "DC" level is not removed, up to two data points are generally bypassed (not used) in any further calculations. A variable, j, can be defined so that the "first valid bin" above zero $\delta\tau$ that is included in calculations corresponds to j = 0.

In order to remove measurement noise from subsequent calculations, $P(\delta\tau)$ is compared to a threshold level, T_1 , typically set to 200 % of the RMS noise level of the detection system. It is now necessary to determine whether the fibre is negligibly or randomly mode coupled.

If it is found that the first X valid points of $P(\delta\tau)$ are all below T_1, this indicates that $P(\delta\tau)$ shall have discrete spike features characteristic of negligibly coupled fibres. The value of X is equal to 3, unless zero-padding is used in the Fourier analysis. In that case, the value of X can be determined from:

$$X = \frac{3 \times \left(number\_original\_data\_points\right)}{total\_length\_of\_array\_after\_zeropadding} \tag{A.5}$$

PMD is calculated using Equation (A.6) for a negligible mode coupling fibre, or PMD is calculated using Equations (A.7) and (A.8) for a random mode coupling fibre.

A.3.3.2 PMD calculation for fibres with negligible mode coupling

For a negligibly coupled fibre (e.g., a high birefringence fibre) or for a birefringent component, such as those used for calibration, $R(\lambda)$ resembles a chirped sine wave (Figure A.3). Fourier transformation will give a $P(\delta\tau)$ output containing a discrete spike at a position corresponding to the relative pulse arrival time, $\delta\tau$, the centroid of which is the PMD value $<\Delta\tau>$.

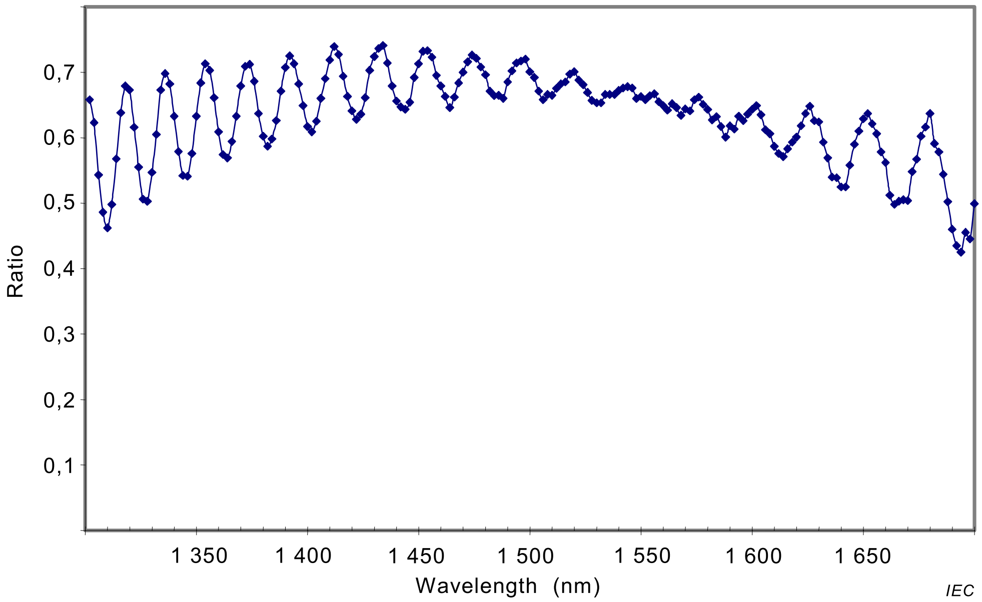


Figure A.3 – A chirped sine wave

To define the spike centroid $<\Delta\tau>$, those points where $P(\delta\tau)$ exceeds a second pre-determined threshold level T_2, typically set to 200 % of the RMS noise level of the detection system, are used in the equation:

$$< \Delta\tau >= \sum\nolimits_{e=0}^{M'} \left[P_e(\delta\tau)\delta\tau_e\right] / \sum\nolimits_{e=0}^{M'} P_e(\delta\tau) \tag{A.6}$$

Therefore, M' + 1 is the number of data points of P within the spike which exceed T_2.

In Equation (A.6), $<\Delta\tau>$ is typically quoted in picoseconds. If no spike is detected (i.e., $M' = 0$), then PMD is zero. Other parameters such as the RMS spike width and/or spike peak value can be reported.

If the device under test contains one or more birefringent elements, more than one spike will be generated. For a number n concatenated fibres/devices, up to $2^{(n-1)}$ spikes will be obtained.

A.3.3.3 PMD calculation for fibres with random mode coupling

In instances of random mode coupling, $R(\lambda)$ becomes a complex waveform similar to Figure A.2, the exact characteristics being based on the actual statistics of the coupling process within the fibre/cable. The Fourier transformed data now becomes a distribution $P(\delta\tau)$ representing the autocorrelation of the probability distribution of light pulse arrival times, $\delta\tau$, in the fibre. Figure A.4 shows the Fourier transform of the result of a random mode coupling measurement.

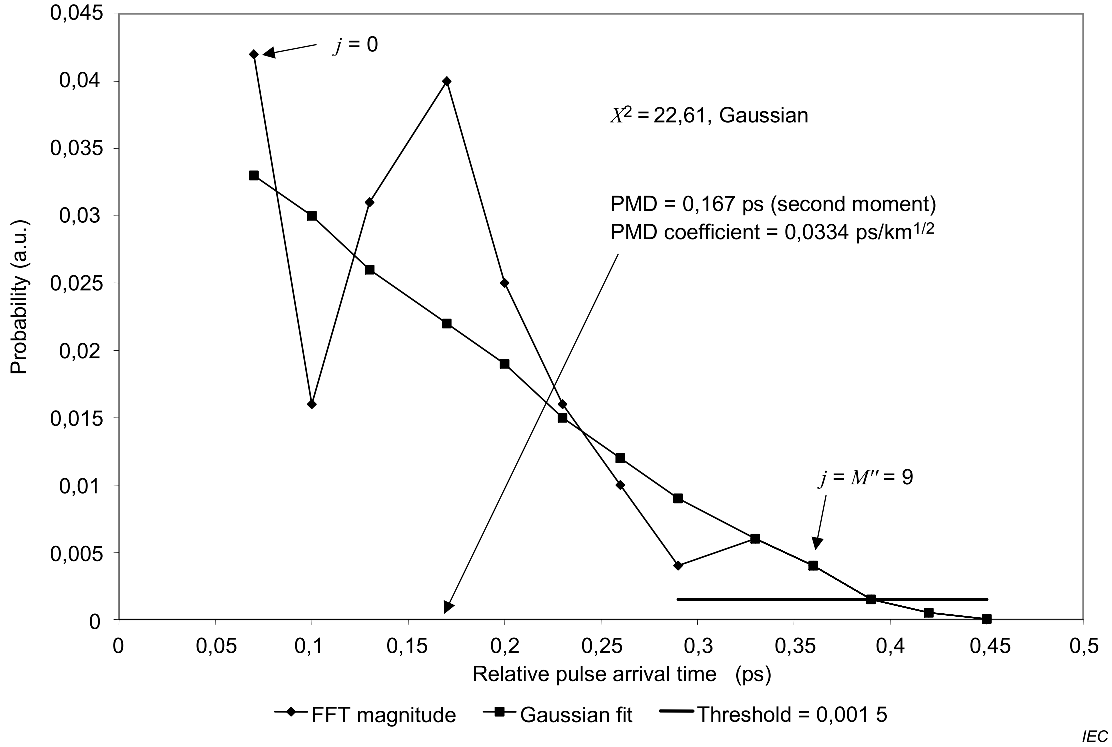


Figure A.4 – PMD by Fourier analysis

Counting up from j = 0, the first point of P is determined which exceeds T_1, and which is followed by at least X data points which fall below T_1. This point represents the last significant point in (i.e., the "end" of) the distribution $P(\delta\tau)$, for a randomly mode-coupled fibre that is not substantially affected by measurement noise. The $\delta\tau$ value for this point is denoted $\delta\tau_{\text{last}}$, and the value of j at $\delta\tau_{\text{last}}$ is denoted M".

The square root of the second moment, σ_{R}, of this distribution defines the fibre PMD_{RMS} $<\Delta\tau^2>^{1/2}$ and is given by:

$$< \Delta\tau^2 >^{1/2} = \sigma_{\text{R}} = \left\{\sum_{j=0}^{M''}[P_j(\delta\tau)\delta\tau_j^2] / \sum_{j=0}^{M''}[P_j(\delta\tau)]\right\}^{\frac{1}{2}} \tag{A.7}$$

When the DGD is assumed to follow a Maxwell distribution, the following gives the relationship of the RMS PMD to the PMD defined by the average:

$$< \Delta\tau > = \left(\frac{8}{3\pi}\right)^{1/2} < \Delta\tau^2 >^{1/2} \tag{A.8}$$

A.3.3.4 PMD calculation for mixed coupling fibre systems

There can be instances where both negligibly coupled fibre/components and randomly coupled fibre(s) are concatenated to form a link. In this case, both centroid determination (Equation (A.6)) and the second moment derivation (Equation (A.7)) may be required. Note that spikes in $P(\delta\tau)$ can only be determined beyond the $\delta\tau_{\text{last}}$ computed.

A.3.4 Spectral range

For randomly coupled fibres, sufficient spectral range shall be used to form the spectral ensemble with sufficient precision. Using the widest possible spectral range (e.g., at least

200 nm) can minimize the statistical uncertainty. The precision required and, therefore, spectral range shall be specified prior to the measurement.

In addition, very low $\delta\tau$ values will give very long periods in $R(\lambda)$, and the spectral range λ_1 to λ_2 should cover at least two complete "cycles", if possible. The spectral range covered defines the smallest $\delta\tau$ value that can be resolved in $P(\delta\tau)$, $\delta\tau_{\text{min}}$:

$$\delta\tau_{\text{min}} = \frac{2\lambda_1\lambda_2}{(\lambda_2 - \lambda_1)c_0} \tag{A.9}$$

where the factor 2 is introduced to allow for the fact that two data points in P at and adjacent to 0 are generally ignored. For example, for λ_1 = 1 270 nm, λ_2 = 1 700 nm, $\delta\tau_{\text{min}}$ = 0,033 ps.

For negligibly coupled high PMD fibres with ratio data $R(\lambda)$ resembling Figure A.2, the requirement for spectral averaging described above can be relaxed, and the spectral range reduced (e.g., λ_2 – λ_1 ~ 30 nm) in order to allow variation of PMD with wavelength to be examined.

Annex B
(normative)

Stokes parameter evaluation method

B.1 Apparatus

B.1.1 Block diagrams

Figure B.1 shows a possible block diagram for Method B, typically used with JME.

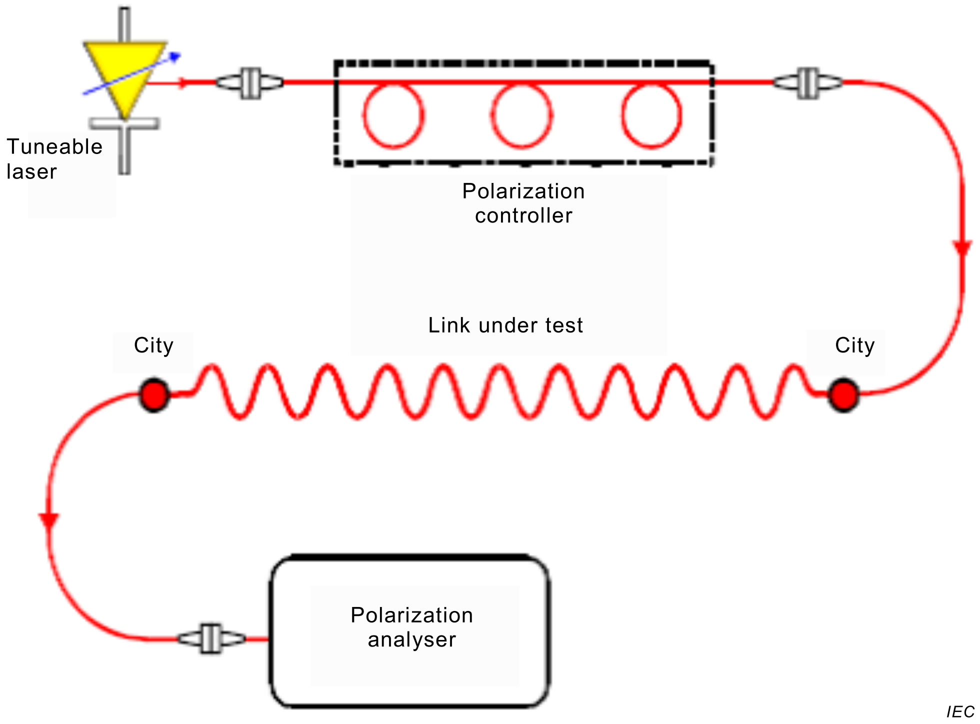


Figure B.1 – Block diagram for Method B using a narrowband (tuneable laser) source

Figure B.2 shows another possible block diagram for Method B, typically used with PSA.

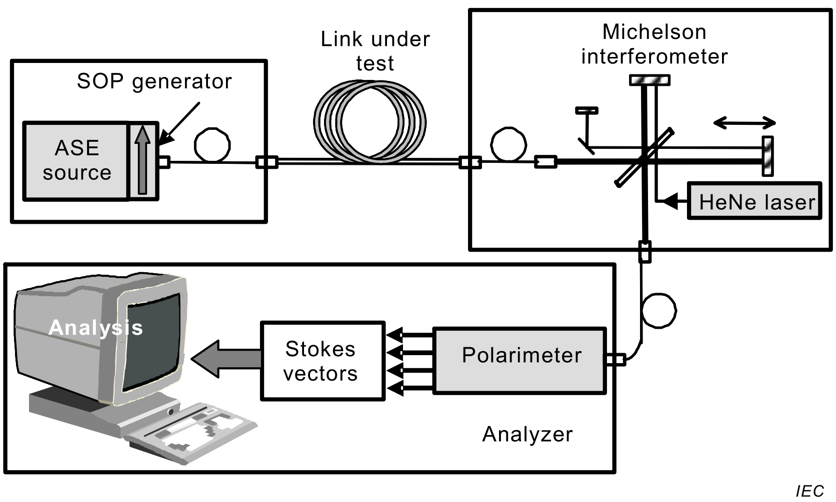


Figure B.2 – Block diagram for Method B using a broadband (ASE) source

B.1.2 Light source

In all cases, two kinds of light sources can be used, depending on the type of polarimeter. A narrowband source such as a tuneable laser shown in Figure B.1 can be used with a polarization analyzer. For amplified links, the DOP of the narrowband source shall be maintained higher than a certain limit throughout the wavelength range to avoid the detrimental effect of the ASE provided by amplifiers. Alternatively, a high power broadband source can be used with a narrow bandpass filtering polarimeter such as an optical spectrum analyzer or an interferometer used as a Fourier transform spectrum analyzer placed before the polarimeter, before or after the link under test. In the case of broadband source, the width of the filter is taken as the spectral width for the purpose of calculations. For amplified links, the broadband source has always a sufficient amount of wavelengths available to avoid the DOP requirement similar to the narrowband source, except if the link ASE is high and spectrally flat over the full wavelength range of interest.

In both cases, the spectral width shall be sufficiently small to maintain the required DOP (see 8.2), but not too small compared to the selected step size to avoid unnecessary coherent interference effects and other spurious noises. In both cases, the wavelength range shall be sufficiently wide to provide a valid PMD measurement of sufficient precision (see B.3).

The polarization controller shall be capable of scanning among at least three known and distinct SOPs (typically linear 0°, 45° and 90° (see Equation (E.4) for each measurement wavelength).

B.1.3 Polarimeter

Use a polarimeter to measure the output Stokes vectors for each selected input SOP and wavelength.

B.2 Procedure

The output of the link is coupled to the analyzer including the polarimeter. The wavelengths are scanned across a range appropriate for the wavelength region and desired precision (see B.3) with a wavelength increment, $\delta\lambda$. For narrowband sources, the wavelength increment is given in terms of the maximum anticipated DGD value, $\Delta\tau_{\mathrm{max}}$, the centre wavelength of the region measured, λ_0, and the speed of light in vacuum, c (299792458 m/s), as:

$$\delta\lambda \leq \frac{\lambda_0^2}{2c\Delta\tau_{\mathrm{max}}} \tag{B.1}$$

For example, the product of maximum DGD and step size shall remain less than 4 ps·nm at 1 550 nm and less than 2,8 ps·nm at 1 300 nm. This requirement ensures that from one test wavelength to the next, the output SOP rotates less than 180° about the principal states axis of the Poincaré sphere. If a rough estimate of $\Delta\tau_{\mathrm{max}}$ cannot be made, a series of sample measurements is performed across the wavelength range, each measurement using a closely spaced pair of wavelengths appropriate to the spectral width and minimum tuning step of the optical source. The maximum DGD measured in this way is multiplied by a safety factor of three. This value is substituted for $\Delta\tau_{\mathrm{max}}$ in the above expression, and the value of $\delta\lambda$ to be used in the actual measurement is computed. If there is concern that the wavelength increment used for a measurement is too large, the measurement can be repeated with smaller wavelength increment. If the shape of the curve of DGD versus wavelength and the mean DGD are essentially unchanged, the original wavelength increment is satisfactory.

For broadband sources, the resolution bandwidth (RBW) of the analyzer shall satisfy the following:

$$RBW_{\lambda} \leq \frac{\lambda_0^2}{5c\Delta\tau_{\max}} \tag{B.2}$$

The measurement data are gathered for each wavelength. For each wavelength, the input SOPs are cycled through the states used (see Equation (E.4)), and the correspondent output Stokes vectors $\hat{H}$, $\hat{V}$ and $\hat{Q}$ are recorded. The output Stokes vectors are normalized to unit length. For the JME calculation, B.3.2, the normalized Stokes vectors shall be converted to normalized Jones vectors using Equation (E.4) (assuming $0 < \theta < \pi$) before the calculation is completed. For the PSA method, B.3.3, the normalized output Stokes vectors are used without conversion.

NOTE Fast scanning systems such as swept wavelength tuneable laser will reduce the impact of fibre movement causing a change in the state of polarization on the measurement. Fast scanning systems can be used successfully on aerial cables or in other installed environments where fibre movement can be observed. [4], [5].

B.3 Calculations

B.3.1 General

The calculation approaches require evaluation of differences between the vectors at one angular optical frequency, ω, and the next at $\omega + \Delta\omega$ (angular optical frequency is given by $\omega = \frac{2\pi c}{\lambda}$). Both calculation approaches result in a series of DGD values versus wavelength. Figure E.4 shows an example of such a function. Alternatively, the DGD values can be displayed as a histogram such as Figure E.5.

The average of these DGD values is reported as the PMD value.

NOTE The JME and PSA calculations are mathematically equivalent for first-order PMD assumptions and where negligible PDL is present.

B.3.2 Jones matrix eigenanalysis (JME)

B.3.2.1 General

For the JME, the response Jones matrix is computed at each wavelength from the Stokes parameters. For each wavelength increment, the product of the Jones matrix $T(\omega+\Delta\omega)$ at the higher optical frequency and the inverse Jones matrix $T^{-1}(\omega)$ at the lower optical frequency are computed. The DGD $\Delta\tau$ for the particular wavelength increment is found from the following argument formula:

$$\Delta\tau = \left| \frac{\mathrm{Arg}\left(\frac{\rho_1}{\rho_2}\right)}{\Delta\omega} \right| \tag{B.3}$$

where

ρ_1 and ρ_1 are the complex eigenvalues of $T(\omega+\Delta\omega)T^{-1}(\omega)$;

Arg denotes the argument function, that is, $\mathrm{Arg}(\eta e^{i\theta}) = \theta$, where η is the magnitude (real) of the complex number;

NOTE $T(\omega+\Delta\omega)$ and $T(\omega)$ are obtained from measurements and calculations of B.3.2.2.

For purposes of data analysis, each DGD value is the DGD at the lower frequency. The series of DGD values obtained from a series of wavelength increments across a wavelength range comprises a single link PMD measurement.

B.3.2.2 Jones matrix eigenanalysis (JME) DGD calculation

R.C. Jones gave an explicit algorithm for experimentally determining the forward transmission Jones matrix T of an unknown linear, time-invariant optical link [2]. The restriction of linearity precludes optical links that generate new optical frequencies. The restriction of time invariance applies only to the polarization transformation caused by the link and does not include the absolute optical phase delay. Therefore, this technique can be used to characterize fibre networks even when the phase delay through the fibre is drifting during the measurement, provided that the DGD measurement apparatus is not affected by phase delay changes.

Measurement of the Jones matrix requires the application of three known and distinct SOPs such as of linearly polarized light to the link. In the process described below, linear SOPs oriented at 0°, 45° and 90° are used, as described by Jones and as illustrated in Figure E.3. The mathematics can be generalized to accommodate other input SOPs.

Any Jones vector V can be completely specified by a magnitude, an absolute phase, and a unit vector $\hat{V}$, which locates the SOP on the Poincaré sphere. To measure the Jones matrix of a link, a stimulus optical field of linear SOP parallel to the x (0°) axis is first generated, and the resulting response unit Jones vector $\hat{H}$ is measured/calculated at the output of the link. Similarly, stimulus fields of linear SOP parallel to the y (90°) axis, and parallel to the bisector of the angle between the positive x and y axes (45°) result in response unit Jones vectors $\hat{V}$ and $\hat{Q}$ respectively.

Three complex ratios independent of the intensities of the three stimulus fields can now be formed from the x and y components of $\hat{H}$, $\hat{V}$ and $\hat{Q}$:

$$k_1 = \hat{h}_x / \hat{h}_y \qquad k_2 = \hat{v}_x / \hat{v}_y \qquad k_3 = \hat{q}_x / \hat{q}_y \tag{B.4}$$

A fourth ratio $k_4 = (k_3 - k_2) / (k_1 - k_3)$ is then found. To within a complex constant β, the transmission Jones matrix T is then given by:

$$T = \begin{bmatrix} k_1 k_4 & k_2 \\ k_4 & 1 \end{bmatrix} \tag{B.5}$$

B.3.2.3 Differential group delay (DGD) determination

The Jones matrix frequency transfer matrix, J, is calculated as:

$$J(\omega) = T(\omega + \Delta\omega) T^{-1}(\omega) \tag{B.6}$$

The DGD $\Delta\tau$ is calculated as:

$$\Delta\tau = |\mathrm{Arg}(\rho_1 / \rho_2) / \Delta\omega| \tag{B.7}$$

where

ρ_1 and ρ_2 are the eigenvalues of $J(\omega)$;

Arg denotes the argument function, that is, Arg $(\alpha e^{i\theta}) = \theta$.

In practice, the impact of PDL on the measurement of $\Delta\tau$ can be reduced by using smaller intervals of radian optical frequency $\Delta\omega$. In all cases, the condition $\Delta\tau\Delta\omega < \pi$ shall be satisfied in order to avoid the ambiguities of the multiple-valued argument function.

B.3.3 Poincaré sphere analysis (PSA) DGD calculation

The analysis presented in [7] is an alternative to the argument function of JME. It is based on an arcsine function.

From the measured normalized Stokes vectors, $\hat{H}$, $\hat{V}$, $\hat{Q}$, compute:

$$\hat{h} = \hat{H}, \qquad \hat{q} = \frac{\hat{H} \times \hat{Q}}{\left|\hat{H} \times \hat{Q}\right|} \times \hat{H}, \qquad \hat{v} = \frac{\hat{q} \times \hat{V}}{\left|\hat{q} \times \hat{V}\right|} \times \hat{q} \tag{B.8}$$

This makes the analysis independent of the input SOPs, and consequently there is no need to know them.

From the Stokes vectors, $\hat{h}$, $\hat{v}$ and $\hat{q}$, form the vector cross products $\hat{c} = \hat{h} \times \hat{q}$ and $\hat{c}' = \hat{q} \times \hat{v}$ at each wavelength. For each wavelength increment, compute the finite differences:

$$\Delta\hat{h} = \hat{h}(\omega + \Delta\omega) - \hat{h}(\omega) \qquad \Delta\hat{q} = \hat{q}(\omega + \Delta\omega) - \hat{q}(\omega) \qquad \Delta\hat{v} = \hat{v}(\omega + \Delta\omega) - \hat{v}(\omega)$$

$$\Delta\hat{c} = \hat{c}(\omega + \Delta\omega) - \hat{c}(\omega) \qquad \Delta\hat{c}' = \hat{c}'(\omega + \Delta\omega) - \hat{c}'(\omega) \tag{B.9}$$

Find the DGD, $\Delta\tau$, for a particular wavelength increment from the following expression:

$$\Delta\tau = \frac{1}{\Delta\omega} \cdot \left[\arcsin\left(\frac{1}{2}\sqrt{\frac{1}{2}\left(\Delta\hat{h}^2 + \Delta\hat{q}^2 + \Delta\hat{c}^2\right)} \right) + \arcsin\left(\frac{1}{2}\sqrt{\frac{1}{2}\left(\Delta\hat{q}^2 + \Delta\hat{v}^2 + \Delta\hat{c}'^2\right)} \right) \right] \tag{B.10}$$

NOTE $\Delta h^2 = \Delta\hat{h} \cdot \Delta\hat{h}$

Each DGD value is taken to represent the DGD at the midpoint of the corresponding wavelength increment.

Annex C
(normative)

Interferometric method

C.1 General

Annex C contains detailed requirements for completing PMD measurements using the interferometric method (INTY) – Method C. This method normally reports the PMD_{RMS} (RMS DGD) metric (see Equation (2)), which can be converted to the PMD_{average} (linear average DGD) metric using Equation (3), under the assumption of a perfect fit with a Maxwell distribution.

There are two versions of Method C with differing measuring set-ups and data analyses:

- the traditional analysis (TINTY) using a basic set-up [8-10];
- a general analysis (GINTY) using a modified set-up compared to TINTY [11].

A generic set-up is shown in Figure C.1, which is the basis of the experimental implementation of Method C (INTY). Variance of this set-up is possible and can be used as shown below.

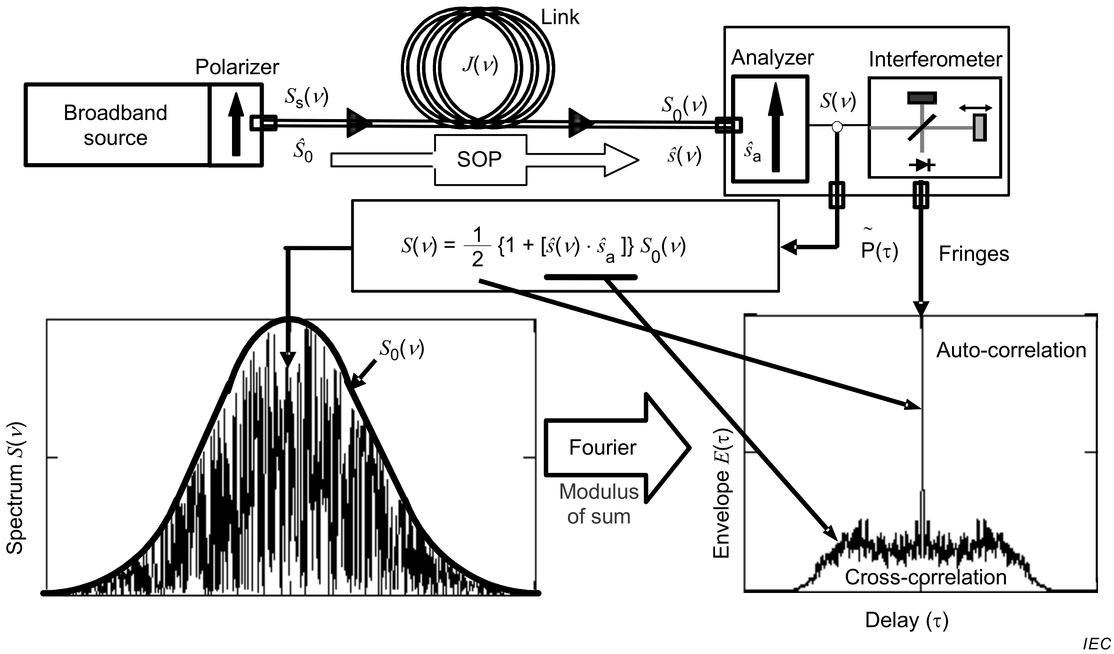


Figure C.1 – Generic set-up for Method C (INTY)

Parameters used in Figure C.1 and throughout Annex C are:

ν optical frequency ($\nu\lambda = c$);

τ difference of round-trip delay between the two arms of the interferometer;

$S_s(\nu)$ optical spectrum, at DUT input ≡ spectral density of $\vec{E}_s(\nu)$, the source spectrum;

$S_0(\nu)$ optical spectrum, at DUT output (analyzer input);

$S(\nu)$ optical spectrum, at analyzer output (interferometer input);

$\hat{s}_0$ input SOP (at DUT input; a unit Stokes vector);

$\hat{s}(\nu)$	output SOP (at DUT output);
$\hat{s}_a(\nu)$	analyzer transmission axis;
$x(\nu) \equiv \hat{s}_A \times \hat{s}(\nu)$	Stokes parameter giving the projection of $\hat{s}(\nu)$ on the analyzer transmission axis. It is this parameter, $x(\nu)$, that contains the PMD information;
$P(\tau)$	optical power at the interferometer output, as a function of delay τ;
$\tilde{P}(\tau)$	dependent part of $P(\tau)$ (AC part);
P_0	constant part of $P(\tau)$ (DC part);
$E(\tau)$	fringe envelope;
$Ex(\tau)$	cross-correlation envelope;
$E_0(\tau)$	auto-correlation envelope.

The optical power at the interferometer output, $P(\tau)$ is equal to the sum of AC and DC parts. Both parts are equal at τ = 0 so the AC part can be calculated. For an ideal interferometer, the AC part is an even function, the right half of which is equal to the cosine Fourier transform of the optical spectrum, $S(\nu)$, emitted from the analyzer. For non-ideal interferometers, some corrections can be applied, depending on the details of the implementation.

For TINTY, the envelope of the interferogram, $E(\tau)$, is the absolute value of the AC part. For GINTY, additional calculations to obtain the cross-correlation and auto-correlation envelopes are described in C.3.2.4. These calculations involve two measured interferograms resulting from the analyzer being set at two orthogonal SOPs.

C.2 Traditional analysis (TINTY)

C.2.1 Apparatus

C.2.1.1 Block diagram

Figure C.2 shows a block diagram for link measurements using Method C (TINTY).

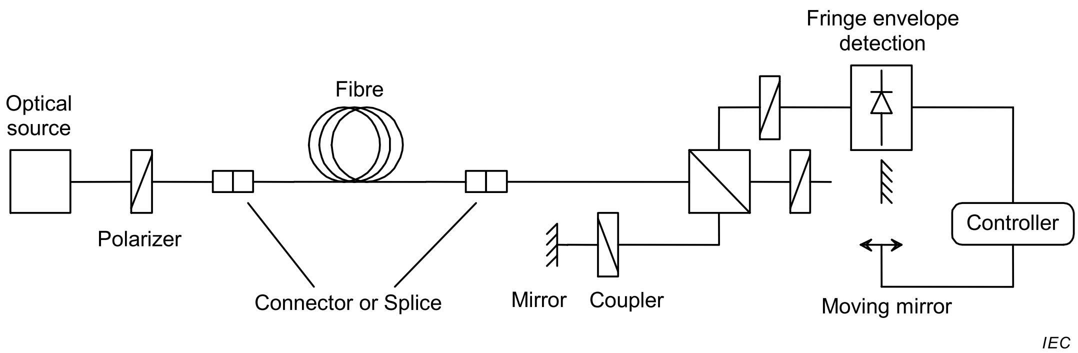


Figure C.2 – Schematic diagram for Method C (TINTY)

C.2.1.2 Light source

A polarized broadband light source is used that emits radiation at the intended measurement wavelengths, such as a LED or a superfluorescent source. The centre wavelength, λ_0, shall be within the 1 310 nm or 1 550 nm windows or any other window of interest. The spectral shape shall be approximately Gaussian, without ripples that could influence the autocorrelation function of the emerging light. The source linewidth (also called spectral width in the LED field), $\Delta\lambda$, shall be known to calculate the coherence time, t_c, which is determined with the following:

$$t_c = \frac{\lambda_0^2}{\Delta\lambda\, c} \tag{C.1}$$

where

λ_0 is the source centre wavelength;

$\Delta\lambda$ is the source linewidth;

c is the speed of light in vacuum.

C.2.1.3 Beam splitter

The beam splitter is used to split the incident polarized light into two components propagating in the arms of the interferometer. The splitter can be an optical fibre coupler or a corner-cube beam splitter.

C.2.1.4 Interferometer

A Michelson type interferometer of either an air type or a fibre type is located at the detector end of the link under test.

C.2.2 Procedure

C.2.2.1 Calibration

The equipment is calibrated by measuring a high birefringent fibre of known PMD. Alternatively, an assembly of high birefringent fibres of known PMD can also be measured.

C.2.2.2 Routine operation

One end of the link is coupled to the polarized output of the light source. The other end is coupled to the interferometer input.

The optical output power of the light source is adjusted to a reference value characteristic for the detection system used. To get a sufficient fringe contrast, the optical power in both arms of the interferometer shall be almost identical.

A first acquisition is made by moving the mirror of the interferometer arm and recording the intensity of the light. From the obtained fringe pattern for one selected SOP, the PMD delay can be calculated as described in C.2.3. A typical example of a fringe pattern for random polarization mode coupling is shown in Figure C.3. The fringe pattern shall be a smooth Gaussian spectrum representative of an ideal random mode coupling.

In case of insufficient mode-coupling, or in case of low PMD, it is recommended that the measurement be repeated for different SOPs or that the polarization state be modulated during the measurement in order to obtain a result which is an average over all the SOPs.

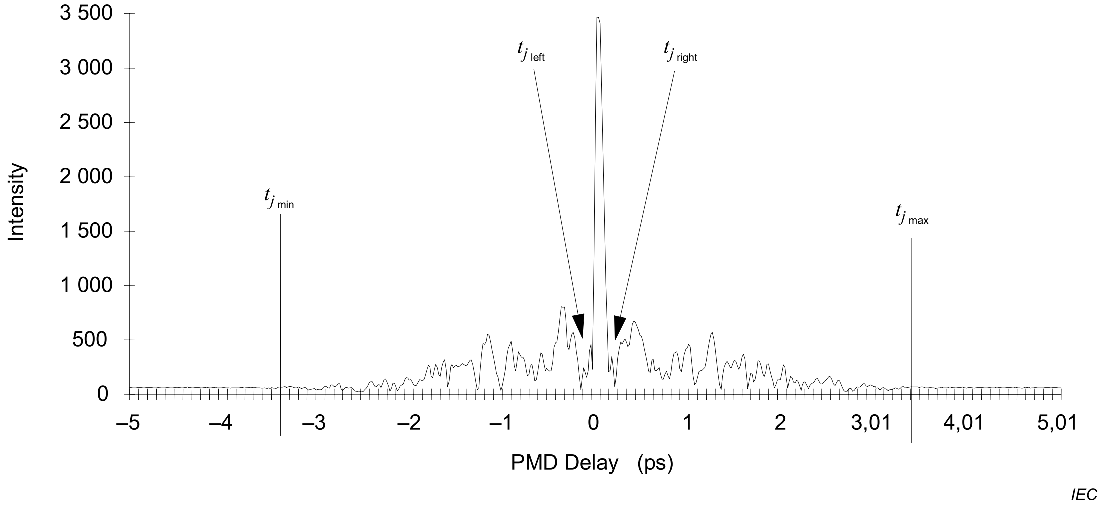


Figure C.3 – Typical data obtained by Method C (TINTY)

C.2.3 Calculations

The following calculations are suitable for the strong/random mode-coupling regime associated with links. The spread in the interferogram – discounting the centre peak – is characterized.

The PMD_{RMS} value is determined from the second moment (RMS width) of the cross-correlation function of the detected signal (interferogram).

$$PMD_{\text{RMS}} = \left(\sqrt{\frac{3}{4}}\right)\sigma_{\varepsilon} \tag{C.2}$$

where σ_{ε} is the RMS width of the cross-correlation envelope. A detailed algorithm for the calculation from a measured fringe envelope is described in Annex H.2.

For certain assumptions given below, Equation (C.2) can be related to Equation (2) as:

$$\left\langle \Delta\tau^2 \right\rangle = \frac{3}{4}\left\langle \sigma_{\varepsilon}^2 \right\rangle \tag{C.3}$$

Equation (C.3) is obtained from the theory [8-10] given the following assumptions (notably):

– ideal random mode coupling;

NOTE 1 Ideal random coupling means $L/h \to \infty$, and a uniformly distributed birefringence axis. L is the device length and h is the polarization coupling-length. For a device consisting of N concatenated birefringent segments of length h, this corresponds to $N \to \infty$ with uniformly distributed axes.

NOTE 2 Analysis of no or negligible mode coupling is possible.

– a purely Gaussian source, with no ripples;

– PMD $>> \sigma_0$

where σ_0 is the RMS width of the auto-correlation envelope;

– ergodic conditions.

NOTE 3 Given that the source is Gaussian, the result will be some form of weighted average of the DGD values. This weighting is not specified in TINTY but is in GINTY. For this reason, Method C (TINTY) is expected to give a different result for a given wavelength range and time than one of the methods that use a rectangular weighting. The assumption of ergodic conditions does result in the validity of the expected value relationship. In practice, the wavelength ranges sampled by different implementations of the other methods will also vary, which would imply getting different results between them as well.

C.3 General analysis (GINTY)

C.3.1 Benefit

GINTY allows some of the assumptions required for Equation (C.3) with TINTY to be removed, notably:

- the assumption of ideal random mode coupling is not required;
- the assumption of a Gaussian source is not required;
- the assumption that the PMD is large compared to the width of the auto-correlation function is not required.

C.3.2 Apparatus

C.3.2.1 Block diagram

In addition to the set-up shown in Figure C.2, Figure C.4 shows a block diagram for link measurements using Method C (GINTY).

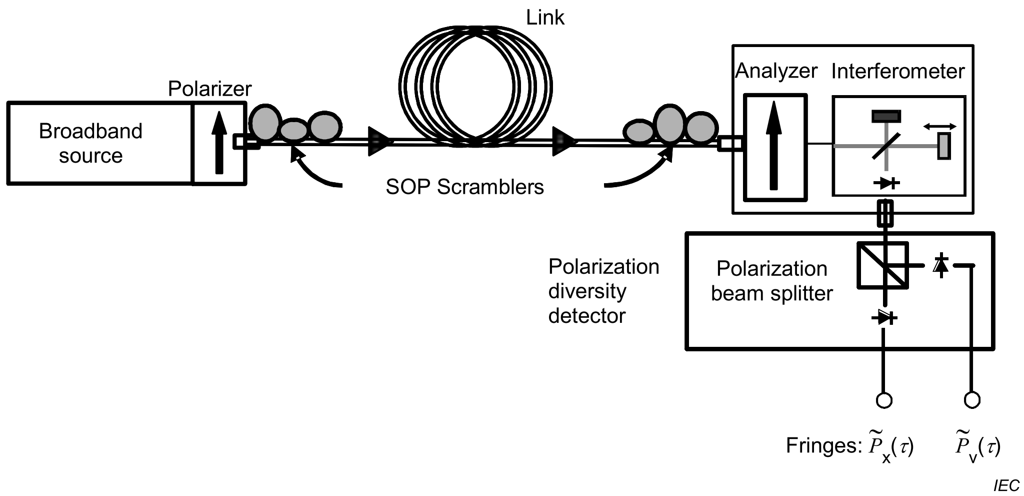


Figure C.4 – Schematic diagram for Method C (GINTY)

When Method C (GINTY) uses the set-up shown in Figure C.4, two interferograms, P_x and P_y, are obtained. These two interferograms correspond to two orthogonal analyzer settings using the setup of Figure C.1. The polarization scramblers allow randomizing both the input SOP and the base of the two effective output analyzer states. The "fringes" shown in Figure C.4 correspond to the AC parts of the idealized interferograms, which could also be obtained from the setup of Figure C.1 for a given pair of input polarizer and base analyzer settings. The sum and difference of these two fringe arrays are squared to obtain the squared envelope of the auto-correlation and cross correlation functions.

C.3.2.2 Light source

A broadband light source is used that emits radiation at the intended measurement wavelengths, such as a LED or a superfluorescent source. The centre wavelength shall be within the 1 310 nm or 1 550 nm windows or any other window of interest. There are no other particular requirements for the source spectrum.

C.3.2.3 Polarizers/scramblers

Polarization scrambling can be used as detailed in 7.6.

C.3.2.4 Polarization beam splitter

A polarization beam splitter (PBS) can be used to obtain interferograms from output SOPs that are orthogonal (opposite on the Poincaré sphere) for the same I/O-SOP combination. These two interferograms allow the calculation of the autocorrelation and cross-correlation as separate functions. Together with the detection system, the PBS forms a polarization diversity detection system. Means other than the PBS can be used to obtain these interferograms from orthogonal output SOPs.

The two fringe patterns observed with two orthogonal analyzer-axes (opposite on the Poincaré sphere) can be used to calculate the auto-correlation and cross-correlation envelopes as:

- auto-correlation envelope for a single I/O-SOP:

$$E_0(\tau) = \left|\tilde{P}_x(\tau) + \tilde{P}_y(\tau)\right| \quad \text{(C.4)}$$

- cross-correlation envelope for a single I/O-SOP:

$$E_x(\tau) = \left|\tilde{P}_x(\tau) - \tilde{P}_y(\tau)\right| \quad \text{(C.5)}$$

C.3.2.5 Interferometer

A Michelson type interferometer of either an air type or a fibre type is located at the detector end of the link under test.

C.3.3 Procedure

C.3.3.1 Calibration

The equipment is calibrated following C.2.2.1.

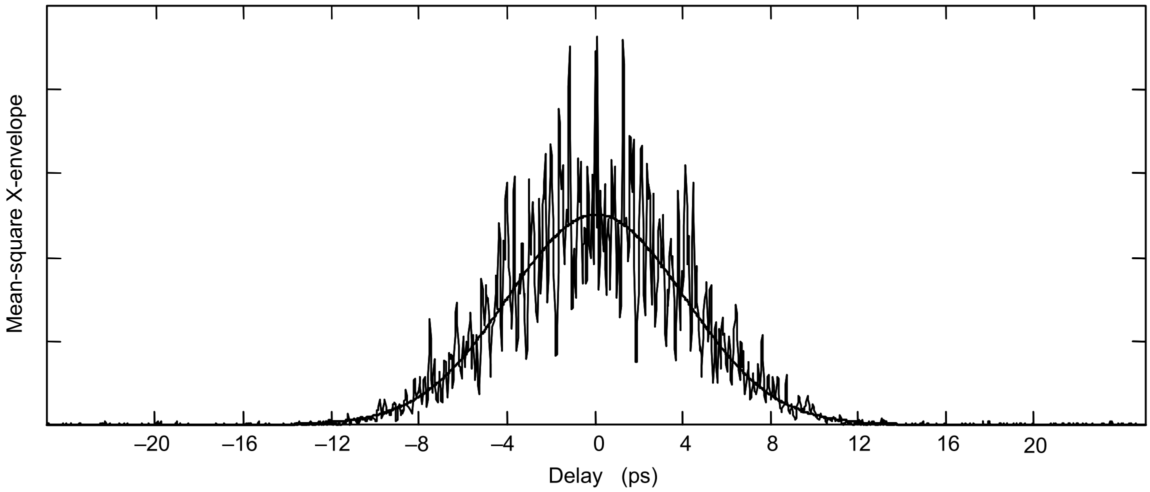
C.3.3.2 Routine operation

In the case of the set-up used as in Figure C.2, one end of the link is coupled to the polarized output of the polarized light source. The other end is coupled to the interferometer input. When the set-up shown in Figure C.4 is used, one end of the link is coupled to the output of the scrambler-polarizer combination. The other end is coupled to the input of the scrambler-interferometer combination.

The optical output power of the light source is adjusted to a reference value characteristic for the detection system used. To get a sufficient fringe contrast, the optical power in both arms of the interferometer shall be almost identical.

A first acquisition is made by moving the mirror of the interferometer arm and recording the intensity of the light. From the obtained fringe pattern for one I/O-SOP combination or from I/O-SOP scrambling, the PMD delay can be calculated as described in C.3.4. Typical examples of mean squared cross-correlation envelopes for random and mixed polarization mode couplings representative of link applications are shown in Figures C.5 and C.6 respectively:

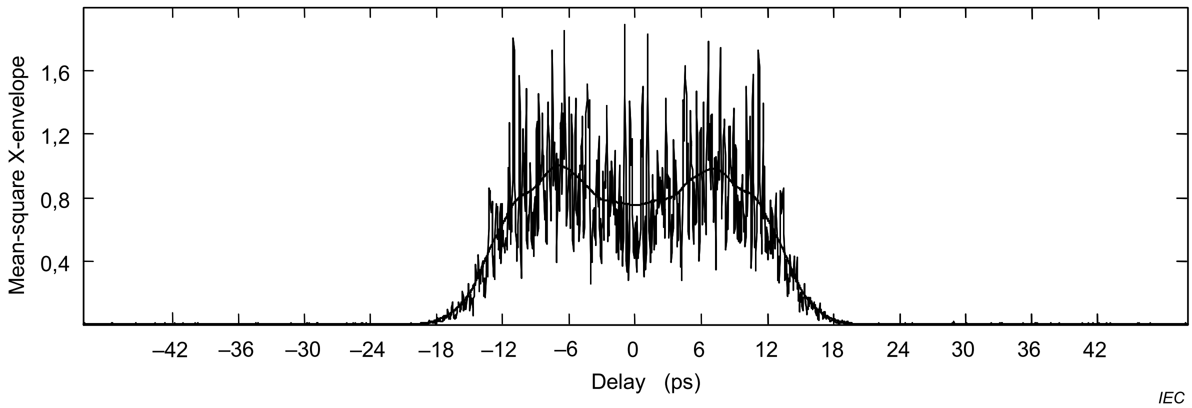
– Figure C.5. Random mode coupling, L/h = 100, PMD = 4,94 ps, σ_A = 50 fs, PMD/σ_A ~ 100



NOTE Smoothing is only used as a visual guide; analysis is neither performed on the smoothing nor any kind of fit.

Figure C.5 – Typical random-mode-coupling data obtained by Method C (GINTY)

– Figure C.6. Mixed mode coupling, nearly-flat envelope; the definition of L/h loses its significance; the DUT is made of 1 random-mode-coupling section with L/h = 10 + 1 negligible-mode-coupling section with $DGD = PMD_{\mathrm{Random}}/4$, PMD = 9,97 ps, σ_A = 50 fs



NOTE Smoothing is only used as a visual guide; analysis is neither performed on the smoothing nor any kind of fit.

Figure C.6 – Typical mixed-mode-coupling data obtained by Method C (GINTY)

C.3.4 Calculations

The following calculations are suitable for any mode-coupling regime associated with links.

The determination of the PMD is based on the squared envelopes of the cross-correlation and auto-correlation interferograms.

First, form the envelopes $E_{0i}(\tau)$ and $E_{xi}(\tau)$ from Equations C.4 and C.5, from N raw interferogram pairs, one for each I/O-SOP combination (indexed with i). Second, form the squared envelopes $E_{0i}^2(\tau)$ and $E_{xi}^2(\tau)$. Third, calculate the mean squared envelopes as:

$$\overline{E}_0^2(\tau) = \frac{1}{N}\sum_i E_{0i}^2(\tau) \qquad \overline{E}_x^2(\tau) = \frac{1}{N}\sum_i E_{xi}^2(\tau) \tag{C.6}$$

Fourth, calculate the RMS width of the two sampled mean squared envelopes σ_0 and σ_x, respectively. A sample algorithm for this calculation is given in Clause H.3. The ideal definitions of these widths are:

$$\sigma_0^2 = \frac{\int_\tau \tau^2 \left\langle E_0^2(\tau) \right\rangle \mathrm{d}\tau}{\int_\tau \left\langle E_0^2(\tau) \right\rangle \mathrm{d}\tau} \tag{C.7}$$

$$\sigma_x^2 = \frac{\int_\tau \tau^2 \left\langle E_x^2(\tau) \right\rangle \mathrm{d}\tau}{\int_\tau \left\langle E_x^2)(\tau) \right\rangle \mathrm{d}\tau} \tag{C.8}$$

The expected value operator in the above equations is with respect to a uniform and random sampling of the I/O-SOPs.

The PMD_{RMS} value that is reported is:

$$PMD_{\mathrm{RMS}} = \left[\frac{3}{2} \left(\sigma_x^2 - \sigma_0^2 \right) \right]^{1/2} \tag{C.9}$$

Equation (C.9) relates to Equation (2) as follows [11]:

$$\frac{\int \Delta\tau^2(\nu) S_0^2(\nu) d\nu}{\int S_0^2(\nu) d\nu} = \frac{3}{2} \left(\sigma_x^2 - \sigma_0^2 \right) \tag{C.10}$$

Using the ideal definitions of the RMS width terms from Equations (C.7) and (C.8), Equation (C.10) is exact for any DGD curve present at the time of measurement and any source spectral characteristic. The left side of Equation (C.10) is a spectrally weighted (by squared power) RMS calculation.

Using the assumption of ergodic conditions:

$$\left\langle \Delta\tau^2 \right\rangle = \left\langle \frac{\int \Delta\tau^2(\nu) S_0^2(\nu) d\nu}{\int S_0^2(\nu) d\nu} \right\rangle \tag{C.11}$$

Annex D
(informative)

Stokes parameter evaluation method using back-reflected light

D.1 Utility

The method in this annex has been shown to yield suitable results for links up to 40 km – without amplifiers [12].

D.2 Apparatus

D.2.1 Block diagram

Figure D.1 shows a layout of the apparatus.

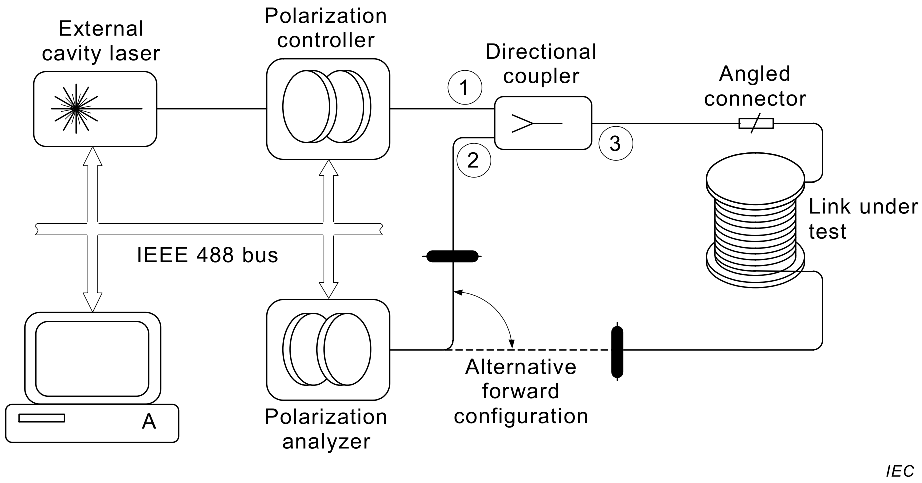


Figure D.1 – Layout for Method D

Most of the elements of the apparatus are the same as for Method B. The following describes the other elements.

D.2.2 Directional coupler

The directional coupler couples the source light into the link and the light that is reflected from the far end of the link into the polarization analyzer. This coupler shall have low PMD (0,1 ps), low PDL (0,1 dB), and low return loss (better than 50 dB) and should be kept stable during the measurement.

D.2.3 Angled connector

This connector shall have return loss of better than 50 dB to avoid mixed signals at the polarization analyzer. The patchcord between the coupler and the angled connector should be kept stable during the measurement.

D.2.4 Far-end termination

The far end of the link shall reflect sufficient light to the polarimeter to detect the output SOPs. A smooth end-cut that is perpendicular to the axis of the fibre may need to be prepared.

D.3 Procedure

The procedure for Method D is the same as for Method B – except that the input end of the link is both input and output for the measurement equipment. In addition, the angled connector at the input end of the link and reflecting end at the output end of the link need special preparation.

D.4 Calculation and interpretation of results

The calculation of DGD at a given wavelength is the same as for Method B. The difference is that the average of these DGD values is not the PMD of the link. Define $<\Delta\tau_B>$ as the average of the back-reflected DGD measurements. The PMD (ps) of the link for the wavelength region measured is then given as:

$$PMD = \frac{2}{\pi}\left\langle \Delta\tau_B \right\rangle \tag{D.1}$$

Annex E
(informative)

Modulation phase-shift method

E.1 Apparatus

E.1.1 Overview and block diagrams

Two primary techniques are available:

- one based on a full search of input SOPs;
- one based on the measurement of the output SOPs associated with a Mueller set of input SOPs.

The full search technique can be implemented with either a simple polarization controller or with a polarization modulator in combination with the controller. Figure E.1 shows a layout for the simple controller, along with alternative connections for controlling the phase reference path. Figure E.2 shows a layout for the polarization modulation case. The apparatus in Figure E.1 is sufficient to complete the Mueller set technique.

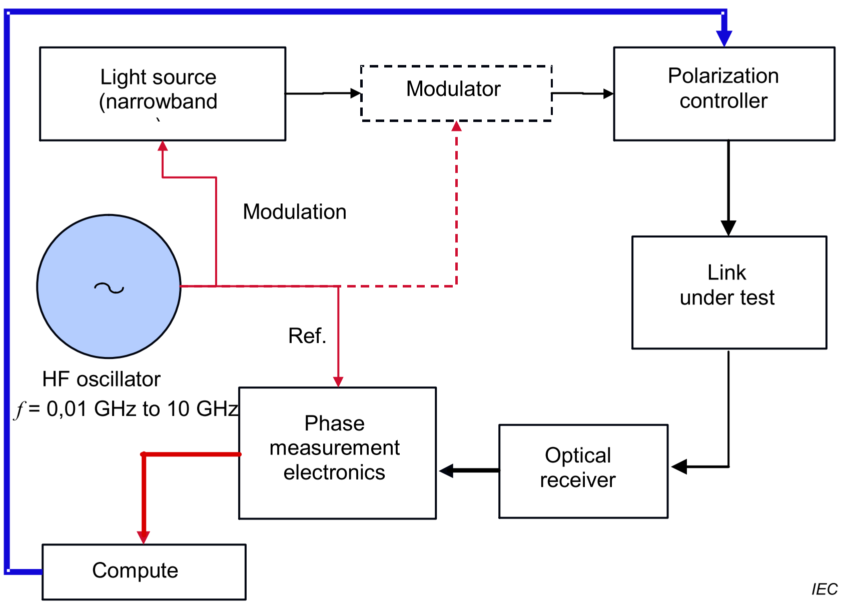


Figure E.1 – Basic apparatus

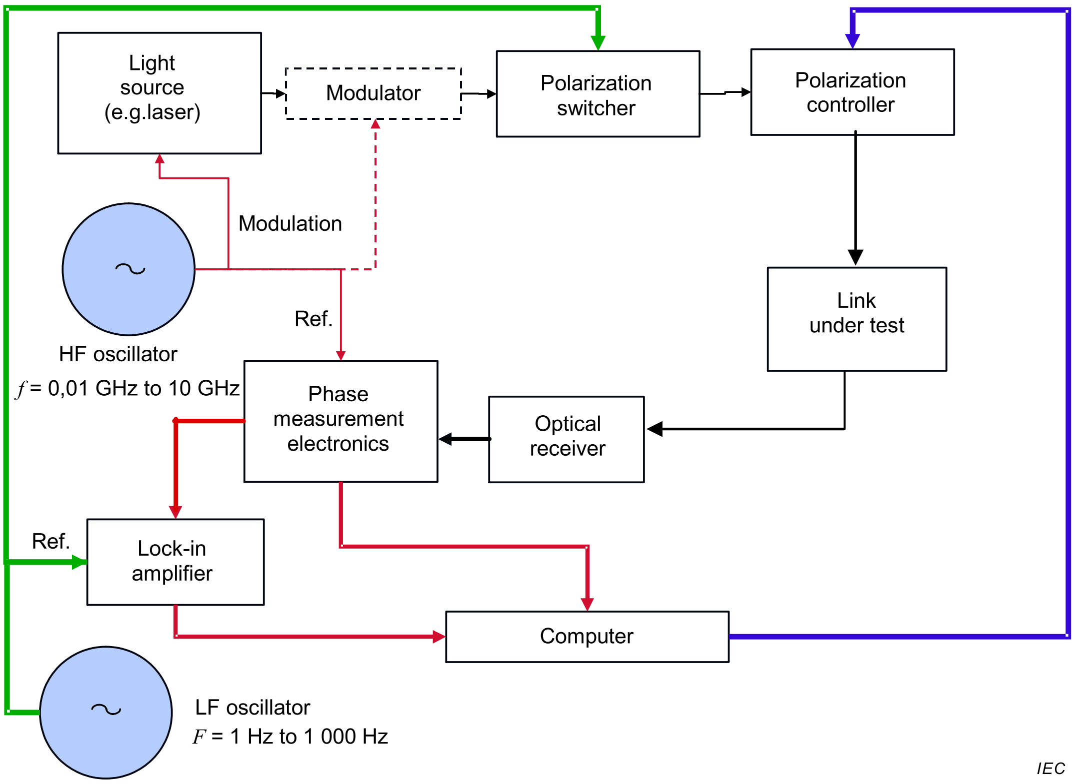


Figure E.2 – Apparatus layout for polarization modulation

E.1.2 Light source(s)

E.1.2.1 General

For the measurement of DGD at each specified wavelength, use multiple laser diodes, tuneable lasers or light-emitting diodes filtered by monochromator or other filter(s). For a "start-stop-measure" wavelength test system, a key issue involved in the selection of the test set is to first determine the required measurement step size based on the DUT spectrum and then to select the source linewidth in accordance with the Nyquist criterion, based on that step size

E.1.2.2 Laser diodes

The laser source spectrum including the centre wavelength and modulated output phase shall be stable over the measurement time period at the bias current, modulation frequency and diode temperature encountered.

Single-longitudinal-mode laser diodes with temperature control and output power stabilization (for example PIN feedback) are typically suitable for use with single-mode optical fibre (SMF).

E.1.2.3 Filtered light emitting diodes

Use one or more light emitting diodes (LED). The LED spectrum, including the centre wavelength and modulation output phase, shall be stable over the measurement time period at the bias current, modulation frequency and LED temperature encountered.

Filter the LED optical spectrum to give a FWHM linewidth in the range 1 nm to 5 nm. A monochromator can be used for filtering or selecting the wavelength.

E.1.2.4 Tuneable diode lasers

Use one or more tuneable diode lasers. The laser spectrum, including the centre wavelength and modulated output phase of each laser at each wavelength used, shall be stable over the measurement time period at the bias current, modulation frequency and diode temperature encountered. Typically completely self-contained temperature-controlled external-cavity laser units can be employed.

E.1.3 Modulation

Modulate the intensity of the light sources to produce a waveform with a single dominant Fourier component. The frequency of the modulation shall be sufficiently high and sufficiently stable to ensure adequate measurement precision.

Modulation can be achieved by direct (internal) current injection to the laser diode or LED. Other (external) forms of modulation means can also be used. Examples are electro-optic modulator devices placed after the laser (see Figure E.1) to modulate the light before it is passed into the link. The modulator shall be sufficiently stable to ensure adequate measurement precision.

A sinusoidal, trapezoidal or square wave modulation is acceptable. A frequency stability of 0,01 ppm is typically sufficient.

It is essential to prevent ambiguities of $2\pi n$ radians (n, an integer) in measuring the phase shift. This can be done by means such as reducing the modulator frequency for large PMD coefficients.

For example, the modulation frequency shall be chosen lower than the frequency that gives a differential phase shift of 2π. This limiting frequency can be estimated as:

$$f_{\mathrm{max}} = \frac{10^3}{DGD_{\mathrm{max}}}\,\mathrm{GHz} \tag{E.1}$$

where DGD_{max} is the maximum expected typical DGD value, in ps.

In fact, for typical DGD values of <100 ps, f_{max} >10 GHz generally exceeds the maximum practical frequencies that can be generated or used.

The modulation at frequency, f, will impart sidebands at $\pm f$ away from the centre wavelength of the source; and in some very narrowband links, this might prove a limitation. To ensure accurate phase measurement, the total occupied bandwidth, including the sidebands of the modulation and the source linewidth itself, shall be less than or equal to the link bandwidth. f is typically chosen as that frequency convenient to phase detection electronics, within any limitations posed by the device bandwidth. f is typically in the range of 10 MHz to 10 GHz.

The RMS repeatability of the DGD measurement, defined here as DGD_{min}, defines a minimum modulation frequency f_{min} that shall be used. Determine f_{min} by the phase resolution $\delta\phi$ of the phase detector:

$$f_{\mathrm{min}} = \frac{\delta\phi \times 10^3}{2\pi \times DGD_{\mathrm{min}}}\,\mathrm{GHz} \tag{E.2}$$

where

DGD_{min} is the RMS DGD repeatability value (ps);

$\delta\phi$ is RMS phase resolution of the phase detector.

E.1.4 Polarization control

The input SOP to the DUT is controlled using a polarization controller. The SOP is selected by means of a control signal from the computer. The computer will select appropriate, known and distinct SOPs, such as orthogonal SOPs, and measure differential delays between these SOPs accordingly.

The polarization controller is used to provide polarized light of a set of SOPs at the input of the DUT. If the light source(s) are not already polarized, the polarization controller shall contain a polarizer at its input. The SOP is controlled by variable birefringent elements following the polarizer. Examples of polarization controllers are liquid crystal retardation plates, loops of birefringent fibre that are mechanically moved, and electro-optic crystal devices, among others.

E.1.5 Input and output optics

Couple the light source output to the link input, and the link output to the detection system input, in a way that ensures that the physical and optical path lengths for each source are constant during the measurement time (this ensures that the differential phase between SOPs does not change because of any change in either path length). Suitable coupling devices can include multichannel single-mode optical switches or demountable optical connectors, but more typically, fusion splices are used. These devices shall be carefully selected in accordance with their lowest possible intrinsic maximum DGD, PDL and group delay over the measurement wavelength range. For calibration fibre measurements, replace "link" with "calibration fibre".

All fibre leads, pigtails and other optical components in the optical path beyond the polarization controller shall exhibit minimal intrinsic maximum DGD, PDL and group delay over the measurement wavelength range. These factors will contribute additional uncertainty in the determination of the DGD of the link.

SOP selection is typically made under computer control, and measurement of the relative phase at the SOPs shall be sufficiently rapid so that the result is not adversely affected by the thermal drift of the delay in the fibre leads.

Scanning the launched SOPs allows the DGD to be determined as the maximum differential phase. The phase detection/computer systems can be implemented in several ways, two examples of which are given below.

In the first example, shown in Figure E.1, the phase detector/computer records the phase at the first test SOP. The polarization controller is set to another SOP. The phase is then recorded at this later SOP. The DGD at the selected wavelength is determined from the maximum difference of the two-phase readings.

In the second example, shown in Figure E.2, a polarization switcher is used to modulate the light SOP at the input to the polarization controller. The polarization switcher can consist of rotating birefringent elements, an electro-optic or photo-elastic modulator, or other elements. The polarization is alternated between two distinct SOPs such as orthogonal at a frequency, F, of several tens of Hz, allowing lock-in detection of a true differential phase output from the phase detector (see E.1.6). This "polarization modulation" allows efficient removal of thermal drift effects. The phase detector produces an AC signal, synchronous with the SOP modulation, with amplitude proportional to the differential phase between the two SOPs. This signal is subsequently demodulated by a lock-in amplifier to produce a DC signal representing the differential phase. The DGD at the selected wavelength is determined from the maximum differential phase from a scan of the launch SOPs over the Poincaré sphere.

E.1.6 Optical detector and phase detection electronics

Use an optical detector that is sensitive over the range of wavelengths to be measured, in conjunction with a phase detector. An amplifier can be used to increase the detection system sensitivity. A typical system might include a PIN photodiode, FET amplifier, and a phase sensitive detector or vector voltmeter.

Optical means, such as a variable optical attenuator or fibre amplifier, can be provided to control the received optical power. The devices are placed in line immediately prior to the detector/receiver. These devices shall be selected carefully in accordance with their lowest possible intrinsic maximum DGD, PDL and group delay over the measurement wavelength range.

The detector-amplifier-phase detection system shall respond only to the fundamental Fourier component of the modulating signal and shall introduce a signal phase shift that is constant over the range of received optical powers encountered.

The phase detection unit/computer will record the phase produced by each SOP and provide an output representing the differential phase/delay between the two SOPs.

E.1.7 Reference signal

Provide the phase detector with a reference signal with the same fundamental Fourier component as the modulating signal against which to measure the differential phases of the signal sources. The reference signal shall be synchronized to the modulating signal and can be derived from the modulating signal.

Examples of reference signals are as follows.

- Where the signal sources and detector are co-located, such as during calibration, an electrical connection can be used between the signal generator and the reference port of the phase detector unit.
- An optical splitter with a detector, inserted before the test specimen, can also be used for co-located equipment. The detector output is amplified, and this signal is used as the reference signal for the phase detector unit.
- The optical modulator at the input can alternatively be controlled from the output end by a separate channel where the electrical modulator is co-located with the phase detection electronics.

E.2 Procedure

E.2.1 Modulation frequency

Set the modulation frequency according to the expected DGD_{max} and DGD_{min} values (see Clause E.1).

E.2.2 Scan the wavelengths and measure DGD

E.2.2.1 Overview

For each technique, there is a method of obtaining the DGD at a particular wavelength. Scan the source across the desired wavelength range and, for each wavelength, measure the DGD as described below.

At the completion of the wavelength scan, the link PMD value is calculated as the average of the DGD values.

E.2.2.2 Full search procedure

For any pair of orthogonal input SOPs, there will be a phase shift difference for a given wavelength. In the context of this measurement, the DGD at the wavelength is derived from the maximum phase shift for any pair of orthogonal input SOPs. The SOP of the pair at which this maximum occurs is known as the PSP.

In general, the orientation of the PSPs of the link will not be known in relation to the polarization axes of the analyzer of the test apparatus. Also, the link PSPs vary with time and wavelength. Other than the uncertainty associated with finding these axes, the main source of uncertainty is the leads within the apparatus itself.

There are several strategies for determining the correct DGD between the PSPs when the PSPs themselves are arbitrary, as described above. In the SOP search methods, a source of uncertainty is the alignment error of the input SOPs with the true principal axes. For example, a 5-degree alignment error will give rise to a 0,4 % uncertainty in the DGD value. Ensure the polarization search alignment error is well controlled.

Examples of suitable implementations are as follows.

- Use the polarization controller in a “search-and-measure” fashion for each test wavelength. Various search algorithms can be adopted to optimize measurement speed. At a given wavelength, use the polarization controller to scan the launch SOP, recording the phase shift for each SOP. Once sufficient SOPs are covered, the maximum and minimum of the recorded phase shifts will correspond to light launched into the PSPs of the link. The DGD at the current wavelength is then proportional to the difference between the maximum and minimum phase shift (see Clause E.3 for the calculation). This difference is referred to as the differential phase shift. The lead birefringence will act on these SOPs in such a way that the net input SOP to the link itself will be aligned to its two PSPs or principal axes. Once the maximum and minimum phases have been detected, the true DGD at this wavelength is the difference in delay between these two SOPs.
- In the case of polarization modulation (Figure E.2), the "search-and-measure" method can be adapted, so that as the polarization controller is scanned, the differential phase shift can be detected directly. This simplifies the search to looking for the maximum differential phase shift, which is proportional to the DGD by the relationship of Equation (E.3).

Designate the maximum phase shift difference for the wavelength as:

$$\Delta\phi_{\mathrm{max}}(\lambda) = \mathrm{max}\left|\Delta\phi_i(\lambda) - \Delta\phi_i'(\lambda)\right| \tag{E.3}$$

where

i is the index and represents the various input SOPs;

$\Delta\phi$, $\Delta\phi'$ are the phase shifts of the input SOPs.

E.2.2.3 Mueller set analysis

A Mueller set of input SOPs is most easily described on the Poincaré sphere, as shown in Figure E.3.

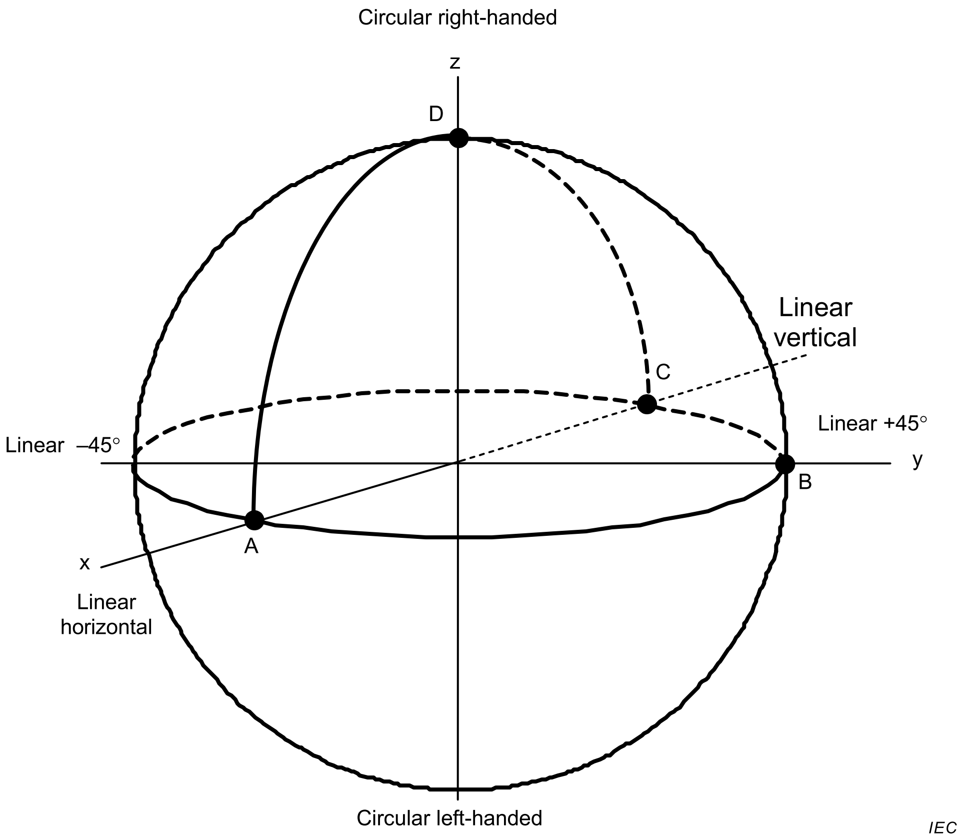


Figure E.3 – Mueller states on Poincaré sphere

SOPs that are orthogonal are 180 ° apart on the Poincaré sphere. Three of the SOPs are on a great circle of the sphere and are inter-mutually separated by 90°, as illustrated in Figure E.3. Using the right hand rule relative to the "north pole", starting at an arbitrary point, A, on the great circle, positions B and C follow by successively adding 90°. Position D is orthogonal to the other points and oriented "up" using the right hand rule. The following Equation (E.4) describes the normalized input Stokes vector, s_0, the parameters of which are used to define an example of a Mueller set where the great circle is on the equator (see Table E.1). The parameter, θ, is the linear orientation of the associated normalized Jones vector, j_0. The parameter, μ, is the phase difference of the x and y elements of that vector.

$$s_0 = \begin{bmatrix} \cos 2\theta \\ \sin 2\theta \cos\mu \\ \sin 2\theta \sin\mu \end{bmatrix} \qquad j_0 = \begin{bmatrix} \cos\theta \exp[-i\mu/2] \\ \sin\theta \exp[i\mu/2] \end{bmatrix} \tag{E.4}$$

Table E.1 – Example of Mueller set

Position	θ	μ	Description
A	0	0	Linear polarization at 0 ° (horizontal)
B	$\pi/4$	0	Linear polarization at 45 ° (45 °)
C	$\pi/2$	0	Linear polarization at 90 ° (vertical)
D	$\pi/4$	$\pi/2$	Circular polarization (spherical)

For each wavelength, cycle the polarization controller through positions A, B, C, and D. For each position, measure the phase shifts (radians), designated, $\Delta\phi_A(\lambda), \Delta\phi_B(\lambda), \Delta\phi_C(\lambda), \Delta\phi_D(\lambda)$, respectively. Calculations using these quantities are given in E.3.

E.2.3 Calibration

Periodically (for example daily or weekly), bring the measurement apparatus together to one location, insert a phase calibration fibre, a two-meter section of low PMD single-mode fibre, into the apparatus and establish a reference signal. For each test wavelength required, repeat the measurement procedure. If it is found that the averaged differential phase is negligible, the contribution associated with the leads and pigtails is also negligible.

E.3 Calculations

E.3.1 DGD calculations

E.3.1.1 Results overview

Both DGD calculation techniques can result in a series of DGD values versus wavelength. Figure E.4 shows an example of such a function. Alternatively, the DGD values can be displayed in a histogram format shown in Figure E.5.

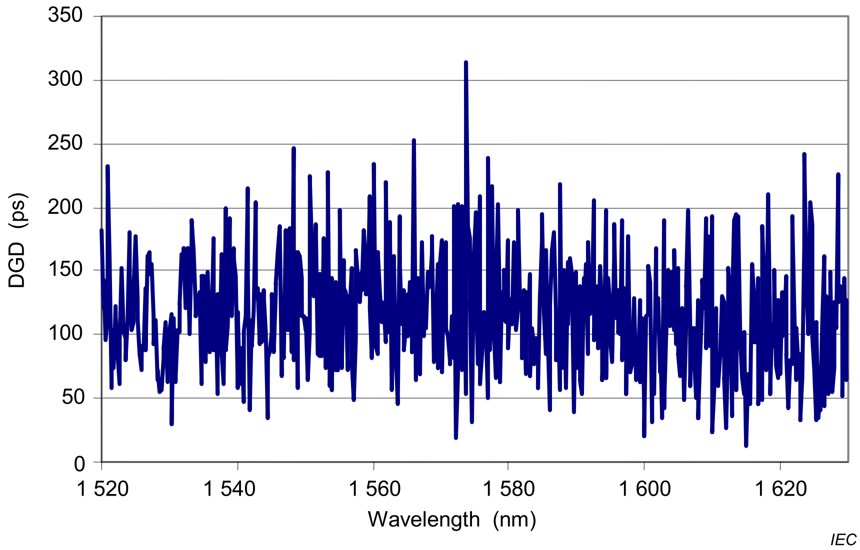


Figure E.4 – DGD versus wavelength

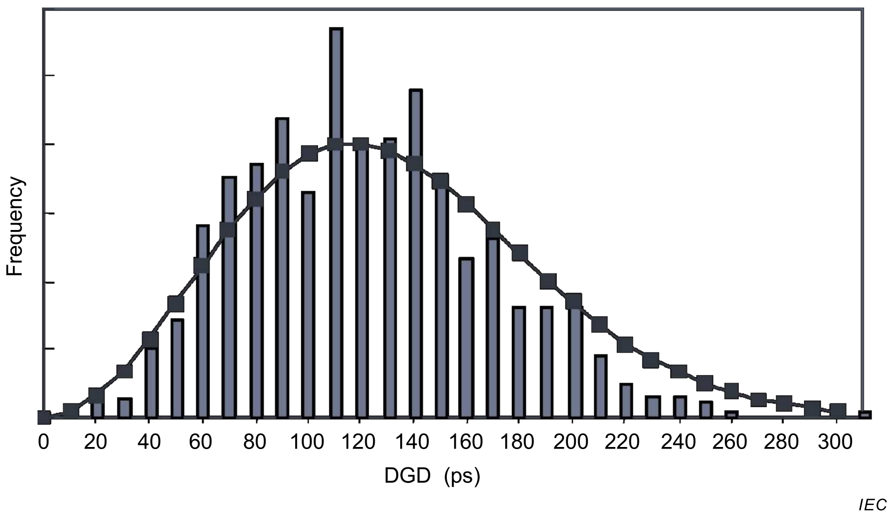


Figure E.5 – DGD in histogram format

E.3.1.2 DGD calculation using full scan technique

Calculate the DGD value (ps), at each wavelength, λ (nm), from the maximum phase difference, $\Delta\phi_{\mathrm{Max}}(\lambda)$ (rad) (see E.2.2.2), and modulation frequency, f, (GHz) as:

$$DGD(\lambda) = \frac{10^3}{2\pi}\frac{\Delta\phi_{\mathrm{Max}}(\lambda)}{f} \tag{E.5}$$

E.3.1.3 DGD calculation using Mueller set technique

Using the notation of E.2.2.3, calculate the average phase of the two PSPs, $\phi_{\mathrm{RF}}(\lambda)$, as:

$$\phi_{\mathrm{RF}}(\lambda) = \frac{\Delta\phi_{\mathrm{A}}(\lambda) + \Delta\phi_{\mathrm{C}}(\lambda)}{2} \tag{E.6}$$

Adjust the measured phase values by the average phase as:

$$\begin{aligned} \phi_{\mathrm{RF,A}}(\lambda) &= \Delta\phi_{\mathrm{A}}(\lambda) - \phi_{\mathrm{RF}}(\lambda) \\ \phi_{\mathrm{RF,B}}(\lambda) &= \Delta\phi_{\mathrm{B}}(\lambda) - \phi_{\mathrm{RF}}(\lambda) \\ \phi_{\mathrm{RF,D}}(\lambda) &= \Delta\phi_{\mathrm{D}}(\lambda) - \phi_{\mathrm{RF}}(\lambda) \end{aligned} \tag{E.7}$$

Calculate the phase difference, $\delta_{\mathrm{RF}}(\lambda)$, as:

$$\delta_{\mathrm{RF}}(\lambda) = 2\arctan\left\{\left[\tan^2\left(\phi_{\mathrm{RF,A}}(\lambda)\right) + \tan^2\left(\phi_{\mathrm{RF,B}}(\lambda)\right) + \tan^2\left(\phi_{\mathrm{RF,D}}(\lambda)\right)\right]^{1/2}\right\} \tag{E.8}$$

The DGD (ps) is calculated using $\delta_{\mathrm{RF}}(\lambda)$ (radians) and the modulation frequency, f, (GHz) as:

$$DGD(\lambda) = 10^3\,\frac{\delta_{\mathrm{RF}}(\lambda)}{2\pi f} \tag{E.9}$$

E.3.2 PMD calculation

The PMD value (ps) for a wavelength range is the average of the DGD values for the range.

Annex F
(informative)

Polarization phase shift method

F.1 Apparatus

F.1.1 Block diagram

The primary technique of the PPS method is given. Figure F.1 shows a block diagram of the PPS method.

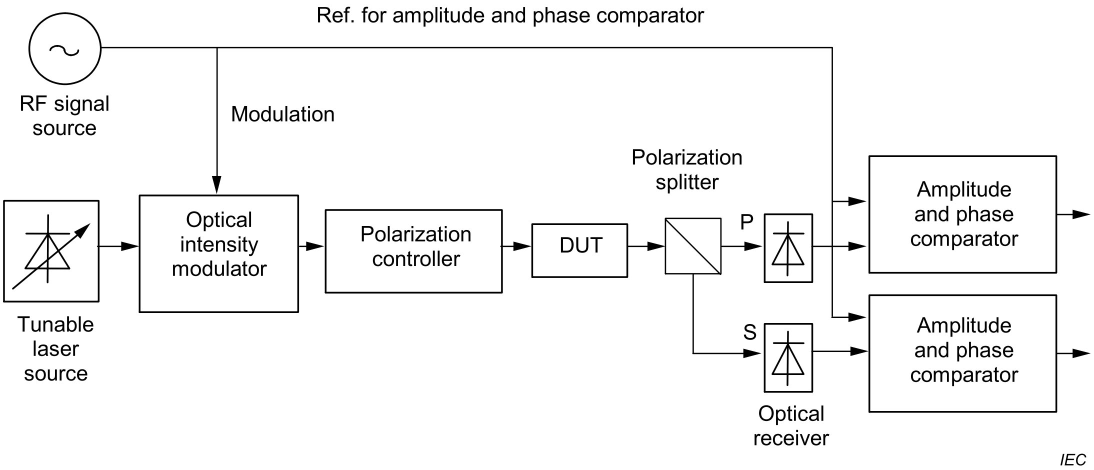


Figure F.1 – Block diagram for Method F (polarization phase shift method)

F.1.2 Light source

A tuneable laser source is used as the light source. The tuning range of the laser shall be sufficient to cover the required PMD wavelength range. The completely self-contained temperature controlled and current controlled external-cavity laser unit is generally employed.

F.1.3 Modulation

F.1.3.1 RF signal source

The RF signal source provides a modulating signal for the optical intensity modulator. Some of the modulating signal is sent to the amplitude and phase comparator as a reference signal. The RF signal source requires a broadband characteristic because it is necessary to provide a sinusoidal modulating signal whose frequency range is typically from 50 MHz to 3 GHz. In the selection of the modulation frequency, undesirable influences of modulation sidebands and the DGD measurement resolution should be considered.

The sidebands are generated on both sides of the optical signal with a frequency difference of f, which is the modulation frequency. This represents the light spectrum spread. The effective wavelength resolution, $\Delta\lambda$ (nm), is restricted by the sideband, and is generally given as:

$$\Delta\lambda = 2 \times \frac{\lambda^2 \times f}{c} \qquad \text{(F.1)}$$

where

λ is the wavelength (nm);

f is the modulation frequency (GHz);

c is the velocity of light in vacuum (m/s).

In addition, the DGD measurement resolution, ΔDGD (ps), is also restricted by the modulation frequency, f, and is typically given as:

$$\Delta DGD = \frac{\Delta\phi \times 10^3}{360 \times f} \tag{F.2}$$

where

$\Delta\phi$ is the phase resolution of the phase comparator (degree);

f is the modulation frequency (GHz).

The total phase accuracy including the frequency stability of the RF signal source is required to be ± 0,3° or less to ensure adequate measurement precision.

F.1.3.2 Optical intensity modulator

The optical intensity modulator receives modulating signals from the RF signal source and modulates the intensity of output light from the tuneable laser source. A LiNbO3 modulator is typically employed.

F.1.4 Polarization control

The polarization controller is used to deliver polarized light of specific SOPs to the DUT. The polarization controller consists of three components: a polarizer, a quarter waveplate, and a half waveplate. Rotating the set of two retardation plates can generate any SOP. The polarization controller needs excellent characteristics to provide expected measurement precision. The angle adjustment resolution is required to be ±0,1°. The polarization extinction ratio is typically required to be 30 dB or more over expected wavelength range.

NOTE One limitation of the method is based on the fact that quarter and half waveplates are generally not spectrally flat.

F.1.5 Output optics

The output from the DUT is coupled into a polarization splitter before the optical detectors. The polarization splitter separates the output from the DUT into two polarized waves, P- and S-polarized light. The polarization splitter consists of a non-isotropic crystal such as a calcite prism and generally possesses a high polarization extinction ratio and very low DGD, group delay and PDL characteristics. The characteristic of the available wavelength range is typically very wide.

F.1.6 Optical detectors

The optical detector converts the modulated light from the DUT into an electrical signal. A PIN photodiode with a good linearity is generally used. The PIN photodiode shall have bandpass characteristics sufficient enough to respond to the modulation frequency of the RF signal source. In addition, to ensure a high signal to noise ratio, a broadband amplifier should be used in the stage after the detector.

F.1.7 Amplitude and phase comparator

The amplitude and phase comparator measures amplitude and phase by comparing the signals for each polarized wave component with the reference modulating signal from the RF signal source. The group delay, τ (ps), is calculated from the phase using the following expression:

$$\tau = \frac{\phi \times 10^3}{360 \times f} \quad \text{(F.3)}$$

where

ϕ is the phase (degree);

f is the modulation frequency (GHz).

F.1.8 Reference signal

The reference signal, which is part of the modulating signal from the RF signal source, is provided to the amplitude and phase comparator. The reference signal shall be synchronized to the modulating signal.

F.2 Procedure

F.2.1 Modulation frequency

The choice of modulation frequency is based on the wavelength resolution, $\Delta\lambda$, required for the measurement results and the DGD measurement resolution, ΔDGD. For more information, refer to F.1.3.1.

F.2.2 Wavelength increment

Two wavelengths are required to obtain a DGD value, because the wavelength differentiation in this wavelength increment, $\delta\lambda$, is used when calculating a DGD. This wavelength increment, $\delta\lambda$, will be called wavelength step size, and the procedure for the determination of $\delta\lambda$ is explained. When the wavelength of the tuneable laser source is changed by the $\delta\lambda$, $\delta\lambda=(\lambda+\delta\lambda)-\lambda$, the polarization angle change of the output SOP from the DUT becomes less than 45°. The $\delta\lambda$ (nm) is usually expressed as:

$$\delta\lambda \leq \frac{\lambda^2 \times 10^3}{4 \cdot c \times \Delta\tau_{\mathrm{max}}} \quad \text{(F.4)}$$

where

λ is the wavelength of the measured range (nm);

c is the velocity of light in vacuum (m/s);

$\Delta\tau_{\mathrm{max}}$ is the maximum anticipated DGD value of the DUT (ps).

For example, the product of maximum DGD value, $\Delta\tau_{\mathrm{max}}$, and wavelength step size, $\delta\lambda$, shall remain less than 2 ps × nm at 1 550 nm.

The calculations are all conducted with the radian frequency increment, $\Delta\omega = \frac{2\pi c\delta\lambda}{\lambda(\lambda + \delta\lambda)}$, which varies with actual wavelength over a range, so measurements made with an equal frequency increment can be used, provided the equivalent expression of Equation (F.4) is met.

F.2.3 Scanning wavelengths and measuring DGDs

The tuneable laser source is used to perform wavelength per wavelength measurement over the desired wavelength range, and the DGD value is calculated at each pair of adjacent wavelengths. The PMD value of the DUT can be calculated after an average DGD has been calculated based on the DGD values previously obtained along the measured wavelength range.

For a given wavelength, this method uses a pair of orthogonal polarized input waves (the 0° and 90° linearly polarized waves). The 0° and 90° linearly polarized waves are launched into the DUT, and the output is separated into two polarized wave components by the polarization splitter. The power and phase for each of the polarized waves (the P- and S- polarized light) at a specific measurement wavelength are measured.

For the 0º input SOP, the output powers and phases are measured for the P and S paths. These are designated as $|T_{11}|^2(k)_{\mathrm{mea}}$ and $\phi_{11}(k)_{\mathrm{mea}}$ for the P output path and $|T_{21}|^2(k)_{\mathrm{mea}}$ and $\phi_{21}(k)_{\mathrm{mea}}$ for the S output path. For the 90º input SOP, power and phase measurements for the two output paths are designated with the same symbols, but with the second subscripts changed from 1 to 2. (The "mea" part of the subscripts is used to distinguish these values from values used in the calibration process.) Given these measured values for a sequence of optical frequencies indexed with k, the first order frequency phase derivatives are calculated according to Equation (F.5).

$$\frac{\Delta\Phi_{ij}(k)_{mea}}{\Delta\omega}=\frac{\Phi_{ij}(k+1)_{mea}-\Phi_{ij}(k)_{mea}}{\Delta\omega} \quad \text{(F.5)}$$

F.2.4 Calibration

A calibration is performed on a low PMD single-mode fibre whose length is 1 m or less before DUT measurement. The phase and power for the two input polarization are measured at both P- and S- detectors for the same wavelengths that will be used in the measurement of the link. The first order phase derivative is also calculated for each pair of adjacent wavelengths. These results follow the same notation as F.2.3, but are subscripted with "cal".

The results for each pair of measurement wavelengths are adjusted by the calibration results as:

$$\left|T_{\mathrm{jl}}\right|^2(k)=\frac{\left|T_{\mathrm{jl}}\right|^2(k)_{\mathrm{mea}}}{\left|T_{11}\right|^2(k)_{\mathrm{cal}}} \qquad \frac{\Delta\Phi_{\mathrm{jl}}}{\Delta\omega}(k)=\frac{\Delta\Phi_{\mathrm{jl}}}{\Delta\omega}(k)_{\mathrm{mea}}-\frac{\Delta\Phi_{11}}{\Delta\omega}(k)_{\mathrm{cal}} \quad \mathrm{jl} = 11 \text{ and } 12 \quad \text{(F.6)}$$

$$\left|T_{\mathrm{mn}}\right|^2(k)=\frac{\left|T_{\mathrm{mn}}\right|^2(k)_{\mathrm{mea}}}{\left|T_{22}\right|^2(k)_{\mathrm{cal}}} \qquad \frac{\Delta\Phi_{\mathrm{mn}}}{\Delta\omega}(k)=\frac{\Delta\Phi_{\mathrm{mn}}}{\Delta\omega}(k)_{\mathrm{mea}}-\frac{\Delta\Phi_{22}}{\Delta\omega}(k)_{\mathrm{cal}} \quad \mathrm{mn} = 21 \text{ and } 22 \quad \text{(F.7)}$$

F.3 Calculations

F.3.1 Results overview

The calculation technique can result in a series DGD values versus wavelength. Figure F.2 shows an example of such characteristics, similar to a link application.

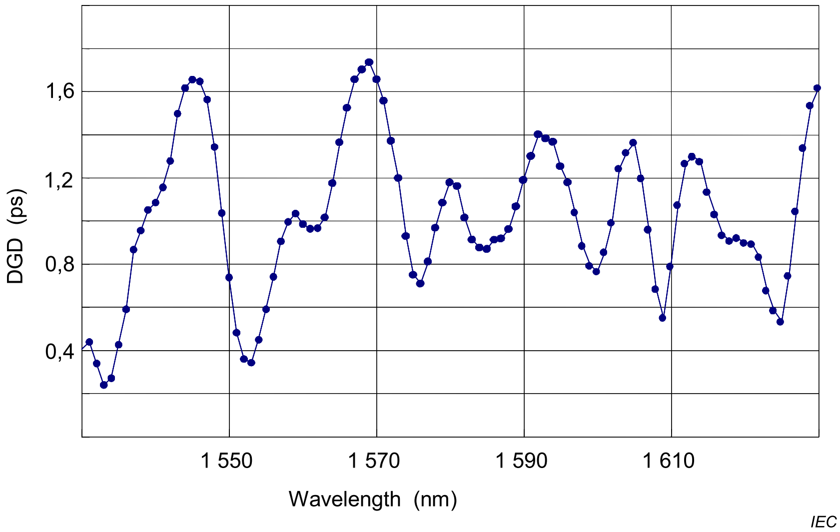


Figure F.2 – DGD versus wavelength for a random mode coupling device

F.3.2 DGD determination

The following parameters are calculated using measured values in F.2.3 and F.2.4 for each pair of wavelengths, indexed with k and k + 1:

$$\begin{aligned}
\bar{\alpha}_1 &= \frac{\Delta\Theta}{\Delta\omega} = \frac{\theta(k+1)-\theta(k)}{\Delta\omega} \\
\bar{\beta}_1 &= \frac{1}{4}\left(\frac{\Delta\Phi_{11}}{\Delta\omega} - \frac{\Delta\Phi_{22}}{\Delta\omega} - \frac{\Delta\Phi_{21}}{\Delta\omega} + \frac{\Delta\Phi_{12}}{\Delta\omega}\right) \\
\bar{\gamma}_1 &= \frac{1}{4}\left(\frac{\Delta\Phi_{11}}{\Delta\omega} - \frac{\Delta\Phi_{22}}{\Delta\omega} + \frac{\Delta\Phi_{21}}{\Delta\omega} - \frac{\Delta\Phi_{12}}{\Delta\omega}\right) \\
\Theta(k) &= \frac{1}{2}\cos^{-1}\left(\frac{|T_{11}|^2 - |T_{21}|^2}{|T_{11}|^2 + |T_{21}|^2}\right) \\
\cos 2\Theta(k) &= \frac{|T_{11}|^2 - |T_{21}|^2}{|T_{11}|^2 + |T_{21}|^2}
\end{aligned} \qquad \text{(F.8)}$$

$$\theta(k) = \frac{1}{2}\cos^{-1}\left(\frac{|T_{11}|^2(k) - |T_{21}|^2(k)}{|T_{11}|^2(k) + |T_{21}|^2(k)}\right) \qquad \text{(F.9)}$$

$$\alpha(k) = \frac{\theta(k+1)-\theta(k)}{\Delta\omega} \qquad \text{(F.10)}$$

$$\beta(k) = \frac{1}{4}\left(\frac{\Delta\Phi_{11}(k)}{\Delta\omega} - \frac{\Delta\Phi_{22}(k)}{\Delta\omega} - \frac{\Delta\Phi_{21}(k)}{\Delta\omega} + \frac{\Delta\Phi_{12}(k)}{\Delta\omega}\right) \qquad \text{(F.11)}$$

$$\gamma(k) = \frac{1}{4}\left(\frac{\Delta\Phi_{11}(k)}{\Delta\omega} - \frac{\Delta\Phi_{22}(k)}{\Delta\omega} + \frac{\Delta\Phi_{21}(k)}{\Delta\omega} - \frac{\Delta\Phi_{12}(k)}{\Delta\omega}\right) \qquad \text{(F.12)}$$

The DGD value for each wavelength is calculated as:

$$DGD(k) = 2\sqrt{\alpha(k)^2 + \beta(k)^2 + \gamma(k)^2 + 2\beta(k)\gamma(k)\cos(2\theta(k))} \tag{F.13}$$

F.3.3 PMD calculation

The PMD value within the measured wavelength range is given by the DGD average value over the measured wavelength range (PMD_{average}).

Annex G
(normative)

Wavelength scanning OTDR and SOP analysis (WSOSA) PMD test method

G.1 General

The WSOSA principle is based on measuring the power reflected from the far end of the link under test. Each datum is determined by recording this reflected power at two closely spaced frequencies, $\nu - {}^{\delta\nu}/_{2}$ and $\nu + {}^{\delta\nu}/_{2}$, for two nominally orthogonal polarizations, x and y, so that four power measurements are made as follows:

$$P_x\left(\nu-\frac{\delta\nu}{2}\right), P_x\left(\nu+\frac{\delta\nu}{2}\right), P_y\left(\nu-\frac{\delta\nu}{2}\right), P_y\left(\nu+\frac{\delta\nu}{2}\right) \tag{G.1}$$

where

ν is the centre frequency of the pair $\nu - {}^{\delta\nu}/_{2}$ and $\nu + {}^{\delta\nu}/2$;

$\delta\nu$ is the frequency spacing between both frequencies of the pair.

NOTE In the procedure, these four measurements are actually repeated, resulting in another four independent power measurements that are distinguished by marking them with a prime mark.

Figure G.1 illustrates the definitions of a frequency pair, its centre frequency and the frequency spacing.

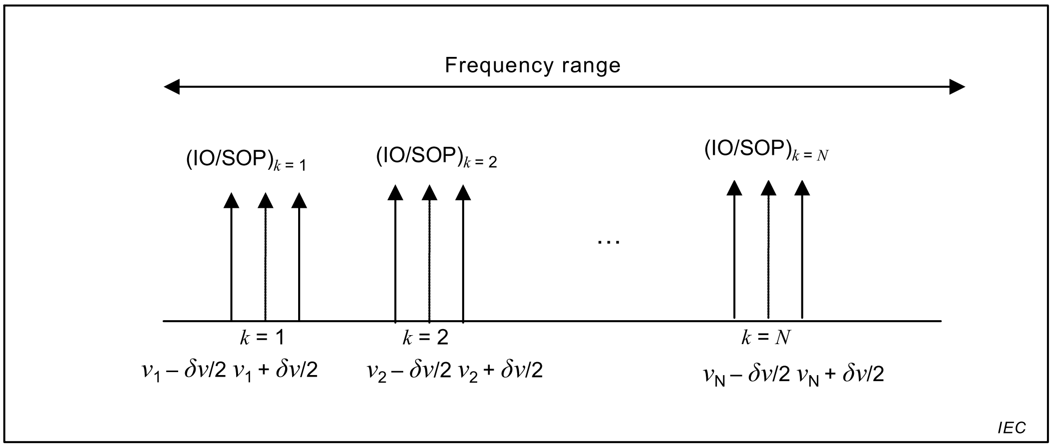


Figure G.1 – Illustration of the frequency domain and parameters for WSOSA

For each pair of frequencies, a random input-output state of polarization (I/O-SOP) is selected. The random I/O-SOPs are selected so each frequency pair is independent of all the other frequency pairs and so the full Poincaré sphere is uniformly covered.

G.2 Apparatus

G.2.1 Block diagram

Figure G.2 shows a schematic of WSOSA generic experimental implementation.

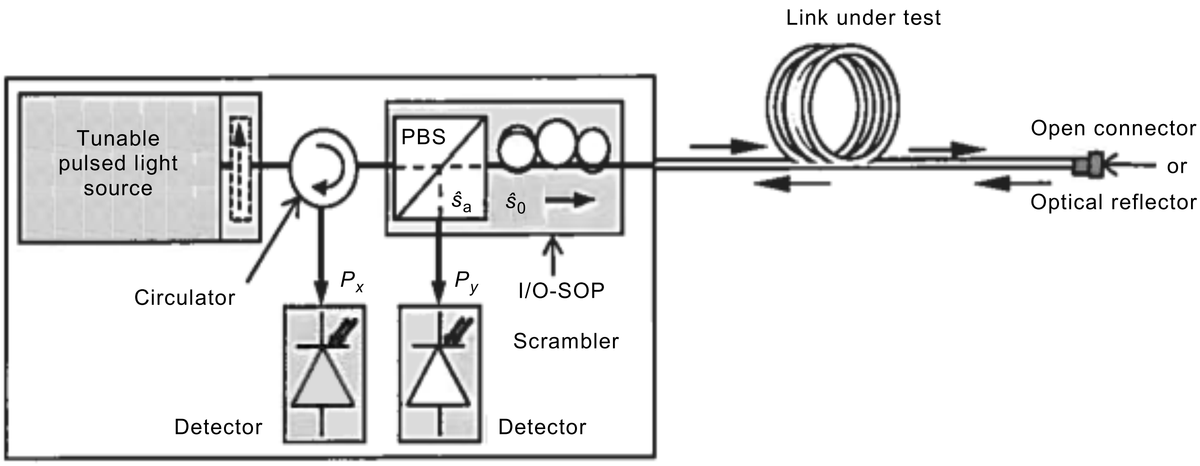


Figure G.2 − Typical generic experimental implementation for WSOSA

Other implementations are possible, such as replacing the polarization beam splitter (PBS) with a polarizer and coupler or by replacing the PBS with a polarizer and eliminating the detector corresponding to P_y altogether.

The parameters used in Figure G.2 and throughout the document are:

P_x, the optical power transmitted on the x side of the PBS,

P_y, the optical power transmitted on the y side of the PBS.

G.2.2 Light source

The light source consists of a relatively narrow-linewidth, widely tuneable-pulsed light source. A wide tuning range should be used; the wider, the better. An example of a typical wide range is 12,8 THz (~100 nm).

The maximum frequency increment, $\delta\nu$, between frequency pairs depends on the round trip PMD_{RMS} of the link. While the method is not very sensitive to this parameter, Equation (G.2) provides a reasonable guideline.

$$PMD_{\mathrm{RMS}}\delta\nu < 0{,}25 \tag{G.2}$$

In theory, any value of the source pulse width can be used, for instance from 10 ns to 20 μs. However, in practice, pulses that are too short will test only over shorter link lengths, unless the link is in fact proportionally short; long pulses that are too long can mix with several reflections; at the limit, this is exactly the case with continuous wave light sources.

The PMD value is independent of the pulse width. If shorter link lengths are tested, equivalent results will be achieved by scaling the pulse duration with the link length or by decreasing the averaging time.

The light source shall be frequency stable to avoid spectral instabilities during the measurement period (this is in fact applicable to every PMD test method).

G.2.3 Launch polarization

The light source should be polarized by the PBS over its full wavelength range. If the source is not naturally polarized, use a polarizer at the source output to meet this requirement. In this case, the polarizer will linearly polarize the source incident light over its full wavelength range in order to fix its SOP. The polarization extinction ratio of the polarizer over the source full wavelength range should be > 20 dB.

G.2.4 Polarization scrambling

Multiple, discrete and random I/O-SOPs are required by WSOSA. Physically, I/O-SOP scrambling consists of inserting a polarization controller, i.e. SOP scrambler, as shown in Figure G.2. Conceptually, the scrambling process can be viewed as follows:

- in the forward direction, the polarized light source SOP followed by the scrambler acts as a single unit, an equivalent polarizer whose axis is set at any point on the Poincaré sphere in order to define the input SOP (I-SOP);
- in the backward (return) direction, the scrambler followed by the analyzer (the PBS) acts as an equivalent analyzer whose axis is set at any point on the Poincaré sphere in order to define the output SOP (O-SOP).

One set of I-SOP/output analyzer-axis (O-SOP) combination is labelled as one I/O-SOP.

The aim is to get the transmissions averaged over uniform, different, discrete and random I/O-SOP distribution. In practice, there exist a number of possible ways to achieve this goal. More details on polarization scrambling and WSOSA can be found in IEC TR 61282-9.

G.2.5 Input/output optics

The input/output optics, as shown in Figure G.2, consist of a circulator, PBS, I/O-SOP scrambler and two detectors, one connected at the output of the circulator and the other to the output of the PBS. The output connector can be an APC connector or a low reflection connector.

A reflector shall be made available at the far end of the link. A high reflection (such as a fibre pigtailed mirror) is preferred in order to maintain sufficiently large dynamic range, but a standard UPC connector with its 4 % reflection (~ –14,7 dB, from an open fibre end) generally allows a good measurement. However, a Faraday-type mirror is not applicable to WSOSA and consequently shall not be used. Reflections from connectors located along the link, as well as the overall Rayleigh backscattering, can be discarded via "time-gated" detection and analysis typical to reflectometry procedure.

G.3 Specimen stability

Minor reflections (such as –50 dB or lower reflectance) located along the last 100 m of the link (for example, with a 1-μs light pulse) will have no effects on the measurements.

The SOPs shall remain stable in the link during the measurement of a frequency pair. Mechanical oscillations and motions of the link during that period can translate into SOP instabilities. The amplitude of these SOP instabilities is a function of the frequency content of the link motion caused by wind (aerial installation), trains running along the link (buried or bridge installation), stress randomly distributed along the link length, or movement of equipment connection cords.

Therefore it is necessary to determine that the link has the required degree of stability before beginning the test.

G.4 Procedure

G.4.1 Set instrument parameters

Some of these settings may need to be modified iteratively depending on the results of trial measurements.

– Select the source frequency region (e.g., 1 300 nm or 1 550 nm) if more than one frequency region is available and set the optical detection on the designated frequency region.

– Inside the designated frequency region, select a wide frequency range, as shown in Figure G.1.

– Use the criterion provided in G.2.2 to define the frequency spacing $\delta\nu$ used during the measurements. As an alternative, determine $\delta\nu$ during a pre-measurement using a smaller number of measurements than during the main measurement.

– Set the acquisition time or number of pulses for power averaging.

  The acquisition time shall be selected carefully, long enough for getting better averaging through noise minimization and improved measurement uncertainty but not too long for getting a long measurement time that is too long and potentially suffering from DUT instabilities. This acquisition time is related to the pulse width and the link roundtrip time (two times the length of the link divided by the velocity of light in the link). For a fixed link length, a shorter pulse width gives better measurement spatial resolution but poorer OSNR and measurement uncertainty. A longer pulse width gives better OSNR but poorer resolution. A compromise is consequently needed between the measurement resolution and the OSNR. 200 ms to 300 ms can be considered as a possible compromise.

– Select the total number of wavelength pairs N.

  Use N ~ 200 as a useful compromise; in this case, the uncertainty $\sigma_{\mathrm{N}=200}$ = 0,075.

  If other values of N than 200 are selected, use the following equation to find σ_{N},

$$\sigma_N = \sigma_{N=200}\sqrt{\frac{200}{N}} \tag{G.3}$$

– Ensure that the I/O-SOP is randomly changed from one centre frequency to another over every frequency pair used in the selected frequency range.

– Attach the link fibre to the instrument.

– Couple the polarization scrambler output to one end of the link.

– If not already done, disconnect the other end of the link or connect the other end of the link to a reflector if required by the targeted measurement dynamic range.

– Ensure that the far end of the link does not consist of an angled connection. If so adapt this angled connection to a flat-face connection.

– In OTDR mode at both frequency extremes, set the sampling location to a position just before the end reflection.

G.4.2 Action of the instrument after measurement initiation

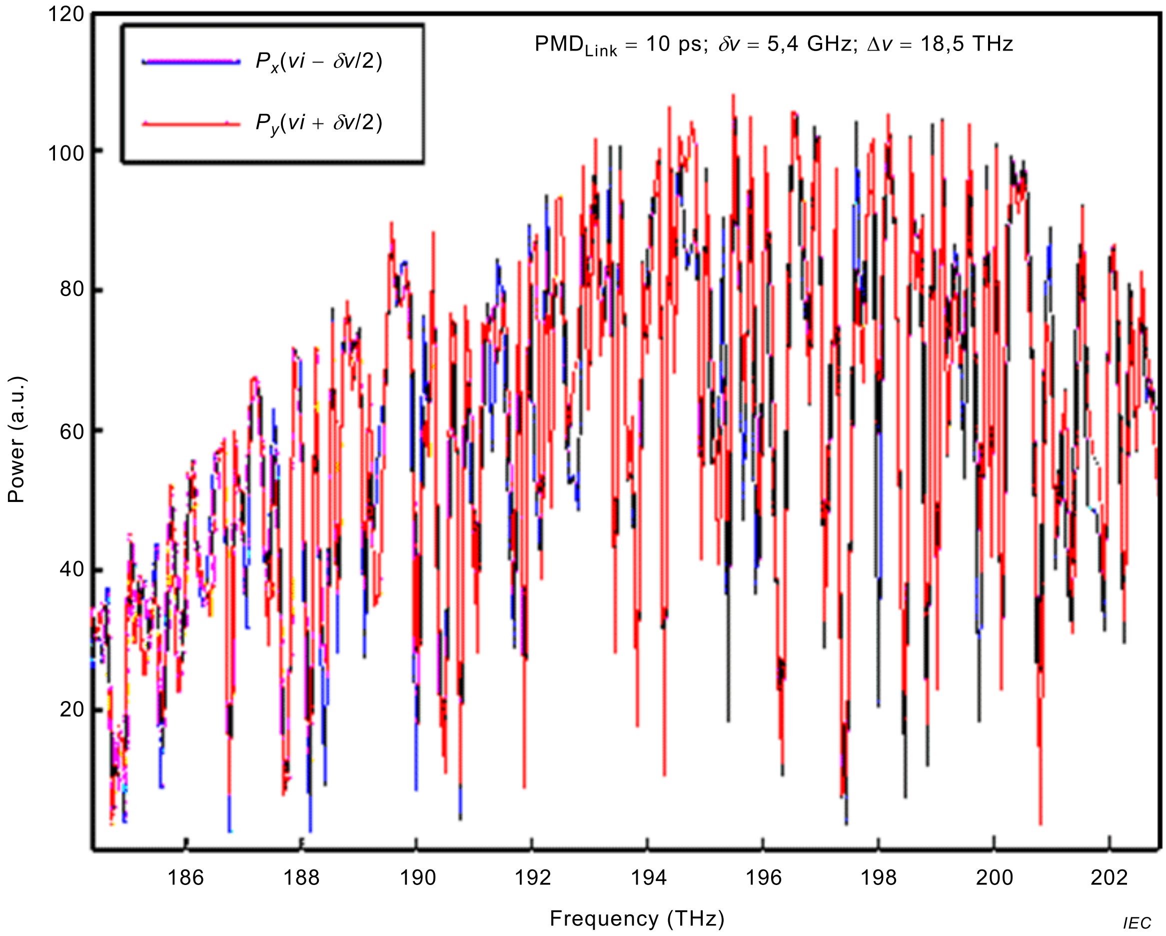
Loop on frequency pairs, indexed by k. For each:

1) Select a random I/O-SOP.
2) Set the tuneable laser to the lower of the two frequencies.
3) Measure the power to detector x, P_x.
4) Measure the power to detector y, P_y.
5) Calculate the normalized transmission, $T_{k-} = \dfrac{P_y}{P_x + P_y}$.
6) Repeat steps 3-5 but record the normalized transmission as T'_{k-}.
7) Set the tuneable laser to the upper of the two frequencies.
8) Repeat steps 3-6 but record the normalized transmission values as T_{k+} and T'_{k+}.

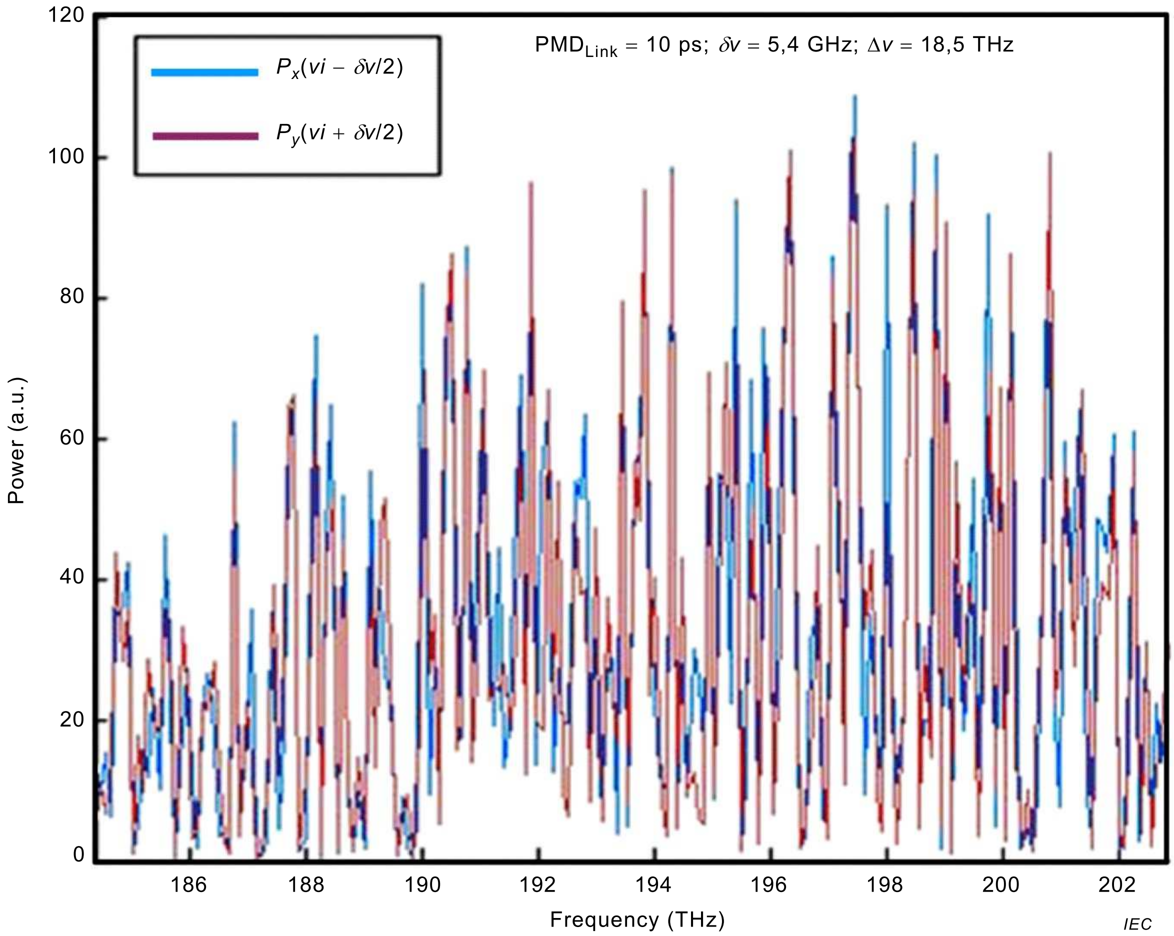
9) Calculate the differences, ΔT_k and $\Delta T_k'$ by subtracting the transmission values of the lower frequency from the values for the upper frequency for the non-primed and primed values, respectively.

NOTE The expression "measure the power" in steps 3 and 4 includes capturing and averaging multiple pulses that are reflected from the location before the end reflection.

Figure G.3 illustrates the power measurements.



(a) Power over a frequency pair at the x-side of the PBS



(b) Power over a frequency pair at the y-side of the PBS

NOTE Each data point in a graph represents one I/O-SOP; a total of 200 random I/O-SOPs were used.

Figure G.3 – Typical power measurement results

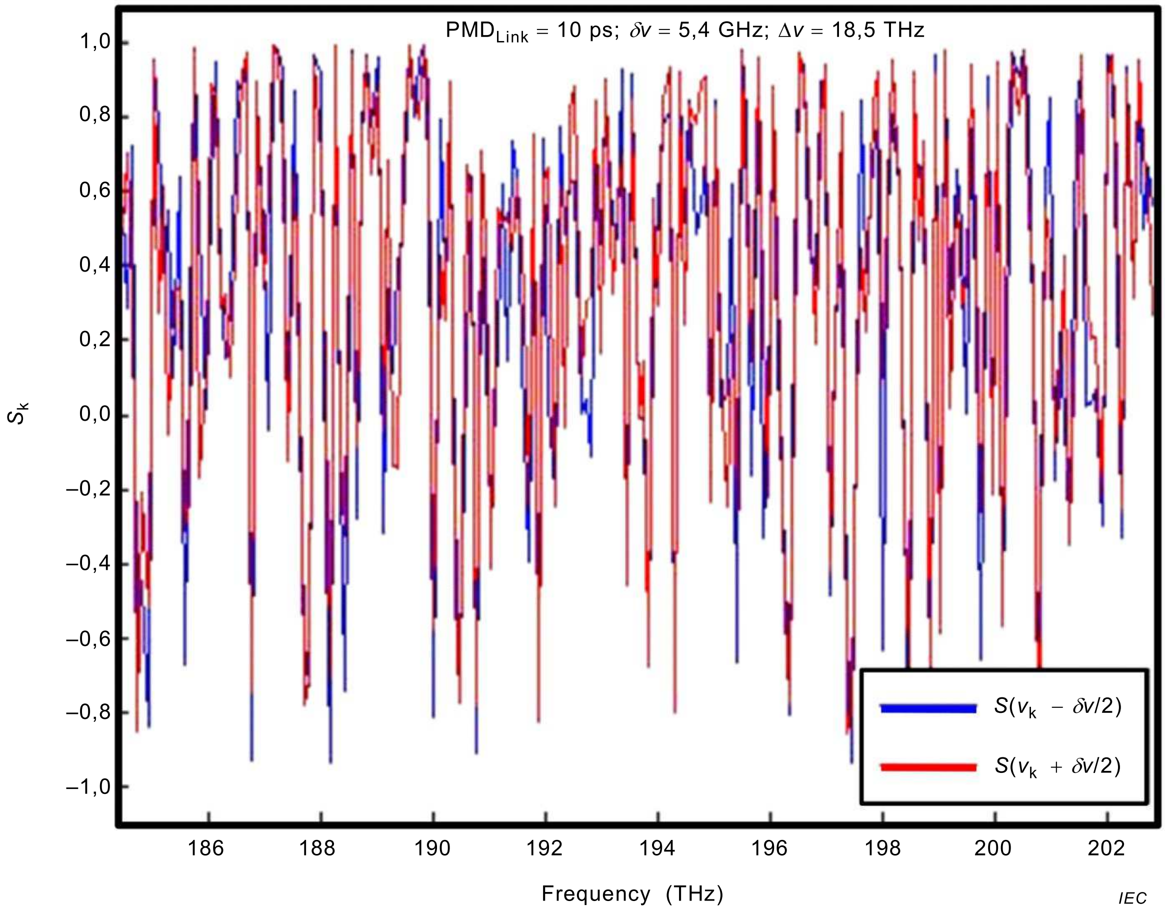
G.5 Calculations

G.5.1 Power normalisation

The power normalization done within the procedure is shown as the following equation.

$$T(\omega)=\frac{1}{2}\left(1+\vec{s}(\omega)\cdot\vec{s}_a\right)=\frac{1}{2}\left(1+s(\omega)\right)=\frac{1}{2}\left(1+\frac{P_x(\omega)-P_y(\omega)}{P_x(\omega)+P_y(\ )}\right)=\frac{P_x(\omega)}{P_x(\omega)+P_y(\omega)} \qquad \text{(G.4)}$$

Figure G.4 illustrates the results for the typical $s(\omega)$ function.



NOTE Each data point in a graph represents one I/O-SOP; a total of 200 random I/O-SOPs were used.

Figure G.4 – Typical $s(\omega)$ for random I/O-SOP

G.5.2 Transmission differences

The transmission differences, ΔT_k and $\Delta T'_k$, calculated in G.4.2 represent two independent estimates of the change in output stokes vector multiplied with the I/O-SOP orientation vector. They are independent in the sense that the noise functions are independent.

Figure G.5 illustrates the transmission difference function.

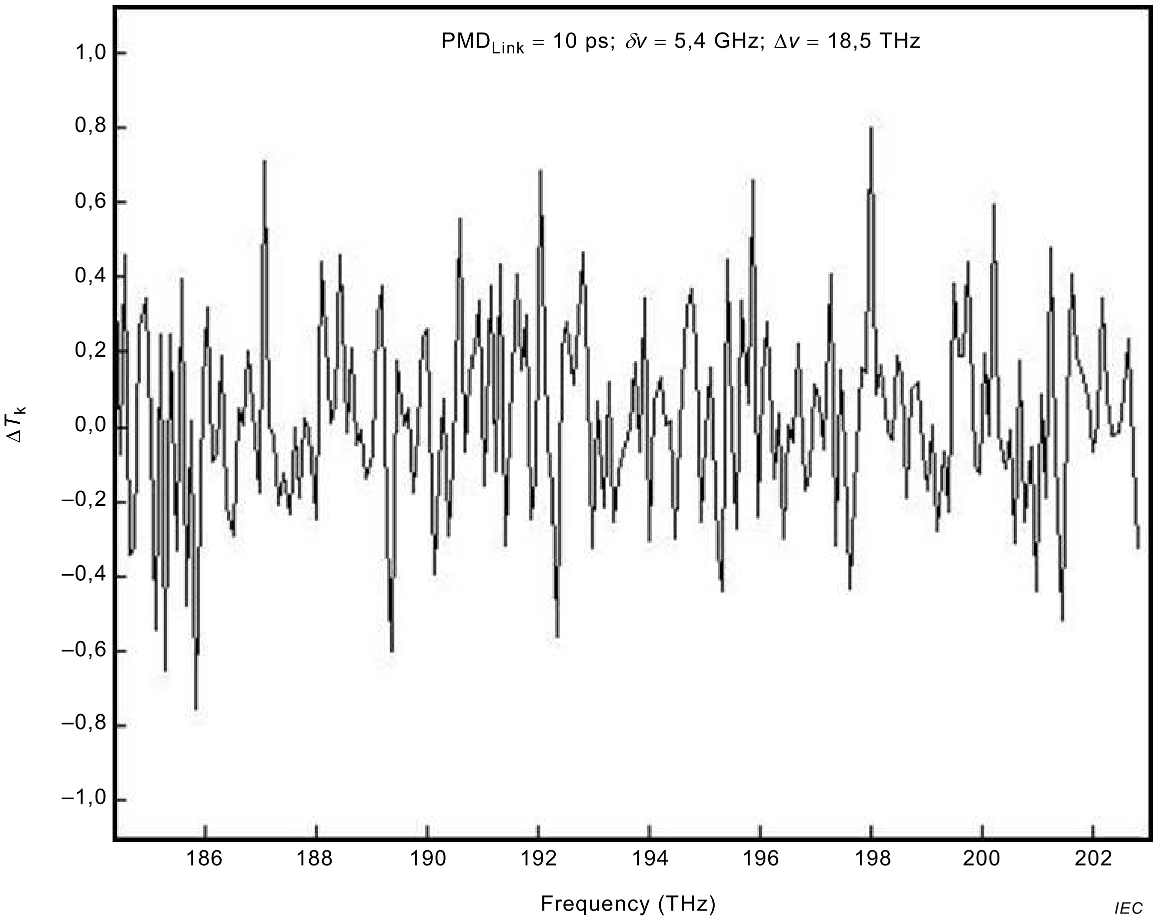


Figure G.5 − Typical transmission difference for a frequency pair and I/O-SOP

G.5.3 Mean-square transmission difference and round trip PMD

Calculate the mean-square transmission difference using the following equation:

$$\left\langle \Delta T^2 \right\rangle = \left\langle \Delta T \Delta T' \right\rangle = \frac{1}{N}\sum_{k=1}^{N} \Delta T_k \Delta T'_k \tag{G.5}$$

Since the primed and non-primed values are independent, the error product term expected value is zero at the same time over a number of occurrences defined by N.

Calculate the roundtrip PMD_{RMS}, as follows:

$$\left\langle \Delta \tau_{RT}^2 \right\rangle^{1/2} = \frac{\alpha_{rs} \Delta T_0}{2\pi\delta\nu}\left[-\ln\left(1 - \frac{\left\langle \Delta T^2 \right\rangle}{\Delta T_0^2} \right) \right] \tag{G.6}$$

where the random I/O-SOP scrambling constant, α_{rs}, and maximum mean-square transmission difference, ΔT_0^2, are given as: $\alpha_{rs}^2 = 15$ and $\Delta T_0^2 = 8/45$

More details on the mathematical development of these equations can be found in IEC TR 61282-9.

G.5.4 Determination of PMD

Calculate PMD, both RMS and mean, as follows:

$$PMD_{RMS}^{2} = \frac{3}{8}\left\langle \Delta\tau_{RT}^{2} \right\rangle \tag{G.7}$$

$$PMD_{MEAN}^{2} = \frac{1}{\pi}\left\langle \Delta\tau_{RT}^{2} \right\rangle \tag{G.8}$$

It is important to note that $\Delta\tau_{RT}$ is not equal to two times the forward or end-to-end $\Delta\tau$ (the DGD parameter of interest). However, when averaged over frequency or time, PMD is related to PMD_{RT} through the average roundtrip factor.

Equation (G.8) depends on an assumption that the DGD distribution is Maxwellian.

G.6 Procedures for measuring PMD on installed aerial links

G.6.1 Instability compensation

In WSOSA, the effect of link instabilities on the measured mean-square transmission difference is characterized by a stability coefficient, g_s, which is a function of the temporal SOP fluctuations in the link, as follows:

$$\left\langle \Delta T \Delta T' \right\rangle_{wI} = \Delta T_0^2 \left[1 - g_s R(\delta\nu)\right] \tag{G.9}$$

Here, the right side is the measured mean-square transmission difference with instabilities present. It is the same as Equation (G.5) but subscripted with "wI" to distinguish it from the instability compensated value, which will be subscripted with "woI". The autocorrelation function, R_x, is the same as that given in IEC TR 61282-9 and is defined as the true function. In the absence of instabilities, the value of the stability coefficient is one.

Instability occurs over time. Two additional difference functions are available from the measured transmission values to calculate the magnitude of instability. These are:

$$\Delta T_{-k} = T_{-k} - T'_{-k} \qquad \Delta T_{+k} = T_{+k} - T'_{+k} \tag{G.10}$$

Equation (G.4) indicates that these differences should be zero in the absence of instability. These are used to define the mean-square transmission instability difference as:

$$\left\langle \Delta T_{-} \Delta T_{+} \right\rangle_I = \frac{1}{N}\sum_{k=1}^{N} \Delta T_{-k} \Delta T_{+k} \tag{G.11}$$

The instability coefficient is defined as:

$$g_s = 1 - \frac{\left\langle \Delta T_{-} \Delta T_{+} \right\rangle}{\Delta T_0^2} \tag{G.12}$$

This can be combined with Equation (G.9) to define a relationship yielding a value for the auto correlation function.

$$R_x(\delta\nu) = 1 - \frac{\langle \Delta T \Delta T' \rangle_{wI} - \langle \Delta T_- \Delta T_+ \rangle_I}{\Delta T_0^2 - \langle \Delta T_- \Delta T_+ \rangle_I} \tag{G.13}$$

Combining this with Equation (G.9) (with g_s set to one), yields the mean-square transmission difference without instabilities.

$$\langle \Delta T \Delta T' \rangle_{woI} = 1 - R_x(\delta\nu) = \frac{\langle \Delta T \Delta T' \rangle_{wI} - \langle \Delta T_- \Delta T_+ \rangle_I}{\Delta T_0^2 - \langle \Delta T_- \Delta T_+ \rangle_I} \tag{G.14}$$

This value should be substituted for $\langle \Delta T^2 \rangle$ in Equation (G.6).

G.6.2 Approaches to reducing the effects of instabilities

G.6.2.1 General

The compensation for instability outlined in G.6.1 has limits. If the instability parameter, g_s, is less than 0,5, consideration for modifying the measurement should be given.

Basically, there are two main options:

- speed up the measurement;
- come back later.

The following sub-clauses include some ideas for speeding up the measurement.

G.6.2.2 Reducing the acquisition time

Reduce the acquisition time for allowing an increase of the I/O-SOP scrambling rate to levels higher than the link self-scrambling rate, for example up to 2,5 rad/s on the Poincaré sphere.

G.6.2.3 OSNR threshold

Increase the minimum SNR threshold by 2 dB to 3 dB by reducing the number of pulses that are averaged for each power measurement.

This approach decreases the measurement sensitivity to the link instabilities, but with a slight decrease of the dynamic range and the spectral range, especially on a high-loss link.

G.6.2.4 Two pulses, each at different frequency

Use a source capable of launching two pulses each at different frequencies, as shown in Figure G.6.

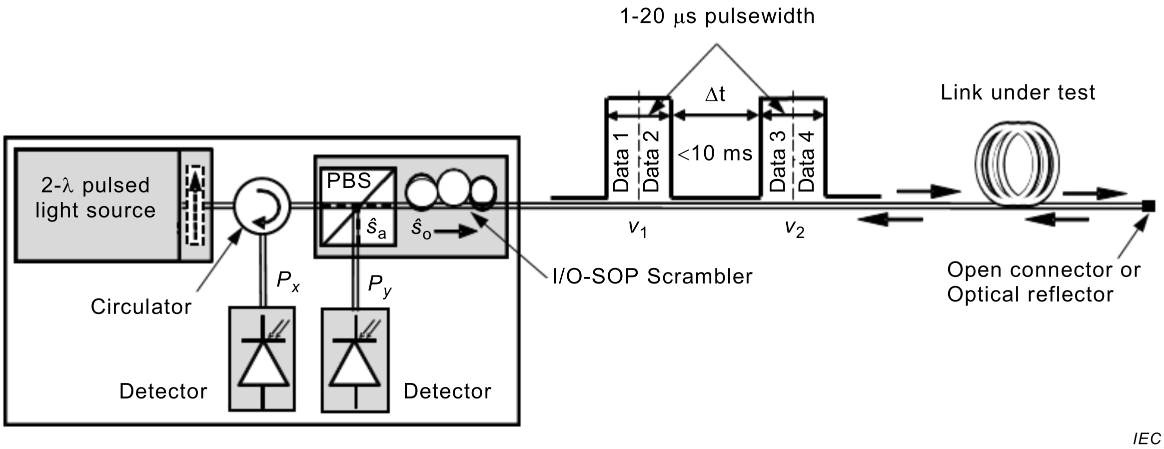


Figure G.6 − Example of 2-pulse implementation in presence of instabilities

This experimental implementation ensures a significant reduction of the effect of the link instabilities on the measured mean-square transmission difference.

G.6.2.5 Two frequencies in the same pulse

Use a source capable of launching two different frequencies in the same pulse with two-channel tuneable filter to separate the two different frequencies.

Annex H
(informative)

PMD determination by Method C

H.1 General

Annex H provides algorithms for determining the RMS width of interferometric data that are measured using Method C. Examples of such interferograms are shown in Annex C. The main reason for these algorithms is to remove the effects of noise.

H.2 Traditional analysis

The following algorithm applies only to Method C (TINTY) in the context of the assumptions described in Annex C.

Let $\tilde{I}_j$ denote the measured intensity of the interferogram at increasing positions t_j (ps), j = 1...N.

Step 1: Computation of the zero intensity $\tilde{I}_0$ and the noise amplitude Na

Definition: N_s *= round (5 N/100) is the integer at which* $\tilde{I}_j$ *is approximately 5 %.*

$$\tilde{I}_0 = \frac{\sum_{j=1}^{N_s}\left(\tilde{I}_j + \tilde{I}_{N_{s-j}}\right)}{2N_s} \tag{H.1}$$

$$X_2 = \frac{\sum_{j=1}^{N_s}\left(\tilde{I}_j^2 + \tilde{I}_{N_{s-j}}^2\right)}{2N_s} \tag{H.2}$$

$$Na = \sqrt{\left(X_2 - \tilde{I}_0^2\right)} \tag{H.3}$$

Step 2: Definition of the shifted intensity I_j

$$I_j = \tilde{I}_j - \tilde{I}_0 \text{ if } \left(\tilde{I}_j - \tilde{I}_0\right) > 4Na \tag{H.4}$$

$$I_j = 0 \text{ if } \left(\tilde{I}_j - \tilde{I}_0\right) \le 4Na \tag{H.5}$$

Step 3: Computation of the centre C of the interferogram

$$C = \frac{\sum_{j=1}^{N_s} t_j I_j}{\sum_{j=1}^{N_s} I_j} \tag{H.6}$$

Step 4: Removal of the autocorrelation peak

Definition:

$$j_l = \text{the largest index } j \text{ such that } C - t_j > t_c \tag{H.7}$$

$$j_r = \text{the smallest index } j \text{ such that } t_j - v > t_c \tag{H.8}$$

where

t_c is the source coherence time.

NOTE For cross-correlation interferograms, the following definition applies:

$$j_r = j_l + 1 \tag{H.9}$$

Step 5: Computation of the second moment S of the interferogram

$$S = \frac{1}{2}\left\{\sqrt{\frac{\sum_{j=1}^{j_l}(t_j - C)^2 I_j}{\sum_{j=1}^{j_l} I_j}} + \sqrt{\frac{\sum_{j=j_r}^{N}(t_j - C)^2 I_j}{\sum_{j=j_r}^{N} I_j}}\right\} \tag{H.10}$$

Step 6: Truncate the interferogram

$$\text{Set } j_{\text{min}} \text{ to the largest index } j \text{ such that } C - t_j > 2S \tag{H.11}$$

$$\text{Set } j_{\text{max}} \text{ to the smallest index } j \text{ such that } t_j - C > 2S \tag{H.12}$$

Step 7: Computation of the second moment σ_s of the interferogram

$$\sigma_s = \frac{1}{2}\left\{\sqrt{\frac{\sum_{j=j_{\text{min}}}^{j_l}(t_j - C)^2 I_j}{\sum_{j=1}^{j_l} I_j}} + \sqrt{\frac{\sum_{j=j_r}^{j_{\text{max}}}(t_j - C)^2 I_j}{\sum_{j=j_r}^{j_{\text{max}}} I_j}}\right\} \tag{H.13}$$

Step 8: Computation of the σ_ε of the Gaussian $\exp\left[-\frac{(t-C)^2}{2\sigma_\varepsilon^2}\right]$ *such that*

$$\sigma_s = \frac{1}{2}\left\{\sqrt{\frac{\int_{t_{j\mathrm{min}}}^{t_{j\mathrm{l}}} (t-C)^2 \exp\left[-\frac{(t-C)^2}{2\sigma_\varepsilon^2}\right]\mathrm{d}t}{\int_{t_{j\mathrm{min}}}^{t_{j\mathrm{l}}} \exp\left[-\frac{(t-C)^2}{2\sigma_\varepsilon^2}\right]\mathrm{d}t}} + \sqrt{\frac{\int_{t_{j\mathrm{r}}}^{t_{j\mathrm{max}}} (t-C)^2 \exp\left[-\frac{(t-C)^2}{2\sigma_\varepsilon^2}\right]\mathrm{d}t}{\int_{t_{j\mathrm{r}}}^{t_{j\mathrm{max}}} \exp\left[-\frac{(t-C)^2}{2\sigma_\varepsilon^2}\right]\mathrm{d}t}}\right\} \quad \text{(H.14)}$$

σ_ε is computed by iteration from (H.14).

H.3 General analysis

The following algorithm results in a robust RMS width for either the composite square cross-correlation or autocorrelation envelopes using Method C (GINTY).

The algorithm is iterative. For a given iteration, the full data array is divided into two sets: the central portion, M, containing the signal and the tails, T, which contain noise. Each iteration results in different definitions for these sets. The result converges when either the computed RMS width ceases to vary or when the set definitions stabilize. For an iteration, the number of data points in each set is denoted as N_M and N_T.

Let $\tilde{I}_j$ denote the measured intensity of the envelope at increasing positions t_j (ps), j = 1...N.

The initial definition of the set T is the first and last 5 % of the whole array.

Step 1: Computation of the zero intensity $\tilde{I}_0$

$$\tilde{I}_0 = \sum_{j \in T} \tilde{I}_j / N_T \quad \text{(H.15)}$$

Step 2: Definition of the shifted intensity I_j

$$I_j = \tilde{I}_j - \tilde{I}_0 \ \text{ all } N \quad \text{(H.16)}$$

Step 3: Computation of the centre C of the interferogram

$$C = \frac{\sum_{j \in M} t_j I_j}{\sum_{j \in M} I_j} \quad \text{(H.17)}$$

Step 4: Computation of the RMS width σ of the squared envelope

$$\sigma^2 = \frac{\sum_{j \in M} (t_j - C)^2 I_j}{\sum_{j \in M} I_j} \quad \text{(H.18)}$$

Step 5: Redefinition of the sets

Define M as the set of points for which $C - 4\sigma \leq t_j \leq C + 4\sigma$.

Define T as the rest of the points.

Step 6: Repetition of steps 1 to 5 until the results converge

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